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Question 1

Not answered

Marked out of 3.00

Schools of Psychology

Match each of the following with the relevant person from the list above;

Founder of the school of individual psychology	Choose...
Founder of operant conditioning model of learning	Choose...
Founder of the school of behaviourism	Choose...

Francis Galton
John Watson
Gustav Theodore Fechner
Johann Weyer
Carl Gustav Jung
Sigmund Freud
Wihelm Greisinger
Alfred Adler
Frederic Skinner

Your answer is incorrect.

Explanation:

Alfred Adler became one of the original four members of Freud's circle in 1902. Adler never accepted the primacy of the libido theory, the sexual origin of a neurosis, or the importance of infantile wishes. Adler thought that aggression was far more critical, specifically in its manifestation as a striving for power, which he believed to be a masculine trait. He introduced the term 'masculine protest' to describe the tendency to move from a passive, feminine role to a masculine, active role. Adler's theories are collectively known as '**individual psychology**'.

The most influential proponent of neobehaviourism since the 1940s was the US psychologist B.F. Skinner, who initiated the study of **operant conditioning**. Skinner's seminal work in operant learning laid much of the groundwork for many current methods of behaviour modification, programmed instruction, and general education.

John Watson, a US psychologist, launched the school of **behaviourism** with the theoretical goal of the 'prediction and control of behaviour'. According to behaviourism, virtually all behaviour can be explained as the product of learning, and all learning consists of conditioning. The contemporary work on classical conditioning of the Russian physiologist Ivan Pavlov, of which Watson was apparently unaware, added further impetus to the behaviourist movement.

Sigmund Freud is credited with the foundation of Psychoanalysis as an intervention in psychiatry though many of his predecessors employed a variety of similar techniques.

Analytical Psychology is a school of psychology based on the concepts of Carl Gustav Jung. Jung gave a fully-developed theory of the structure and dynamics of the psyche, and a detailed theory of personality types.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 1078, 1088.

Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Founder of the school of individual psychology → Alfred Adler, Founder of operant conditioning model of learning → Frederic Skinner, Founder of the school of behaviourism → John Watson

Question 2

Not answered

Marked out of 3.00

Books and Authors

Match each of the proponents below with the book authored from the given list

The Myth of Mental Illness	Choose...
The Divided Self	Choose...
A Secure Base	Choose...

Battie
Freud
Linehan
Pinel
Bleuler
Kraepelin
Laing
Foucault
Szasz
Bowbly
Frankl

Your answer is incorrect.

Explanation:

The *Myth Of Mental Illness: Foundations of a Theory of Personal Conduct* is a controversial anti-psychiatry book by Thomas Szasz published in 1961. Szasz contended that only physical illnesses are real and that mental diseases are 'counterfeit and metaphorical illnesses'.

The Divided Self: An Existential Study in Sanity and Madness was written by the Glasgow-born psychiatrist RD Laing, who promoted a non-treatment approach to mental illnesses including psychosis. Laing's stated aim in his book is to 'make the process of going mad understandable' and his unifying theme is that of ontological insecurity which he believed is central to schizoid and schizophrenic experience.

John Bowlby is the author of *A Secure Base* and is generally considered the founder of attachment theory. His theory of the mother-infant bond was firmly rooted in biology and drew extensively from ethology and evolutionary theory.

Ref: Benning TB. No such thing as mental illness? Critical reflections on the major ideas and legacy of Thomas Szasz. BJPsych bulletin. 2016 Dec;40(6):292-5.

Robertson S. On Laing's The Divided Self. The British Journal of Psychiatry. 2014;204(1):68.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 1080.

The correct answer is: The Myth of Mental Illness → Szasz, The Divided Self → Laing, A Secure Base → Bowlby

Question 3

Not answered

Marked out of 3.00

Delusion syndromes

Identify the psychopathological term used for each of the following:

A woman telephoned the police as she believed her husband was an imposter

A woman walking down the street recognized a stranger as her husband

A 76-year-old woman with Alzheimers dementia has a false belief that some guests are living in her house

Mirror sign
Doppelganger
Intermetamorphosis
Capgras syndrome
Phantom Boarder Syndrome
Folie a deux
Fregoli syndrome

Your answer is incorrect.

Explanation:

Capgras syndrome is a type of delusional misidentification in which the patient believes that a person known to them has been replaced by a 'double' who is to all external appearances identical but is not the 'real person'. This syndrome can occur both in psychiatric disorders, mainly psychotic disorders (especially schizophrenia) as well as major affective disorders, and neurocognitive disorders (e.g., post-traumatic encephalopathy, epilepsy, cerebrovascular disease, dementia, and delirium/acute confusional state).

Fregoli syndrome is a type of delusional misidentification in which the patient believes that strangers have been replaced with familiar people.

Phantom boarder syndrome is a belief that people are in the patient's house although no one is there, as sometimes seen in dementia.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 104, 111.

Karakasi MV et al. In fear of the most loved ones. A comprehensive review on Capgras misidentification phenomenon and case report involving attempted murder under Capgras syndrome in a relapse of a schizophrenia spectrum disorder. Journal of forensic and legal medicine. 2019 Aug 1;66:8-24.

Nomura K et al. Classification of delusions in Alzheimer's disease and their neural correlates. Psychogeriatrics. 2012 Sep;12(3):200-10.

The correct answer is: A woman telephoned the police as she believed her husband was an imposter → Capgras syndrome, A woman walking down the street recognized a stranger as her husband → Fregoli syndrome, A 76-year-old woman with Alzheimers dementia has a false belief that some guests are living in her house → Phantom Boarder syndrome

Question 4

Not answered

Marked out of 5.00

Defence mechanisms

Identify the defence mechanisms used in each of the following clinical situations;

A man who was reprimanded by his supervisor at work picks a fight with his wife at home over a trivial cause

A hostage victim falls in love with the perpetrator

A man with numerous difficulties at work spends most evenings playing squash and finds that this helps his temper

A 50-year-old woman seems unable to acknowledge her recent diagnosis of breast cancer

An 18-year-old man who was abandoned by his parents later joined a humanitarian organization that helps orphans

Passive aggression
Reaction formation
Displacement
Repression
Intellectualisation
Identification with aggressor
Altruism
Sublimation
Denial
Regression

Choose...

Your answer is incorrect.

Explanation:

Displacement is the process by which interest and/or emotion is shifted from one object onto another less threatening one. For example, the person who is having problems at work may displace the anger felt for their boss onto their family by displaying irritability and moodiness at home.

Identification with the aggressor is observed where the victim begins to assume the qualities or faults of the opponent. This may show itself as the battered partner believing she deserves to be beaten and justifying her abuser's aggression to her. The 'Stockholm syndrome' is another example.

Sublimation is a mature defence mechanism that involves expressing unacceptable internal impulses in socially acceptable ways.

Denial is defined as the expressed refusal to recognise a threatening reality. It is of relevance especially to those with serious physical illnesses, where the patient denies being told of the presence of illness in themselves or their loved ones. It may persist despite constant reiteration of the facts.

Altruism describes the mechanism of satisfying one's own needs through the lives of others. It involves vicarious but constructive and instinctually gratifying service to others, even to the detriment of the self.

Ref: Casey P, Kelly B. Fish's clinical pathology: Signs and symptoms in psychiatry, 4th Edition. 2019. p. 129.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 893.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 1067.

The correct answer is: A man who was reprimanded by his supervisor at work picks a fight with his wife at home over a trivial cause → Displacement, A hostage victim falls in love with the perpetrator → Identification with the aggressor, A man with numerous difficulties at work spends most evenings playing squash and finds that this helps his temper → Sublimation, A 50-year-old woman seems unable to acknowledge her recent diagnosis of breast cancer → Denial, An 18-year-old man who was abandoned by his parents later joined a humanitarian organization that helps orphans → Altruism

Question 5

Not answered

Marked out of 4.00

Disorders and Freudian defences

For each of the following disorders given below, identify the defences commonly used.

Paranoid delusions	Choose...
Fugue	Choose...
Dissocial personality	Choose...
Phobia	Choose...

Idealisation
Acting out
Projection
Regression
Denigration
Displacement
Denial
Sublimation
Splitting
Dissociation
Isolation

Your answer is incorrect.

Explanation:

Projection is characterised as perceiving and reacting to unacceptable inner impulses and their derivatives as though they were outside the self. On a psychotic level, this takes the form of frank delusions about external reality, usually persecutory, and includes both perceptions of one's own feelings and those of another with subsequent acting on the perception (psychotic paranoid delusions).

Dissociation is a temporary but drastic modification of character or sense of personal identity to avoid emotional distress. It includes fugue states and conversion reactions.

In acting out, patients directly express unconscious wishes or conflicts through action to avoid being conscious of either the accompanying idea or the affect. Tantrums, apparently motiveless assaults, and pleasureless promiscuity are typical examples.

In psychoanalysis, displacement is a defense mechanism involving redirection of emotions from their original object to a substitute one by a chain of associations. The substitute object may be less threatening that the original one, and the displacement may therefore have the effect of avoiding or reducing anxiety, as seen in phobias.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 1066-1067.

Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Paranoid delusions → Projection, Fugue → Dissociation, Dissocial personality → Acting out, Phobia → Displacement

Question 6

Not answered

Marked out of 5.00

Interview techniques

For each of the following statements given below, identify the interview technique used.

Symptom exaggeration
Clarification
Redirection
Confrontation
Symptom expectation
Interpretation
Summation
Question rephrasing
Limit setting
Facilitation

You seem very upset when talking about your girlfriend. Are you having any problems with your relationship? Choose...

The psychiatrist uses approval nods, leaning forwards slightly to express interest, using responses such as 'Yes. And then?' 'Yeah, go on..', 'Uh-huh', etc.

You mentioned having taken overdoses since your last admission. How many times have you taken overdoses in total? Seven...? Eight...?

What sort of recreational drugs do you usually use when you are drinking?

The patient said 'It is not good if one's parents are divorced even before one goes to school.' The doctor replied 'I'd like to hear more about your parents, but first let me get a picture of what's happening to you of late.'

Your answer is incorrect.

Explanation: Interpretation is clarifying certain associations or relationships that the patient may not see. It is a sophisticated technique and should be used only after the doctor has established some rapport. Should be stated as a hypothesis after sufficient collection of evidence from the interview.

Facilitation is helping patients continue the interview by providing both verbal and nonverbal encouragement. It helps patient to feel that the doctor is listening actively. Encourages flow of information.

Symptom exaggeration: When deception or minimisation is expected, overstating a guessed frequency in order to elicit a true answer. It is also helpful in reducing guilt to a certain extent as the patient feels that the doctor has expected a higher amount of problem that what she/he has brought.

Symptom expectation: Without a formal admission from the patient, asking about details of problem behaviour. The doctor assumes (rightly) that the patient is involved in the act.

Redirection is gently reorienting patient towards the topic of discussion. The motive is to keep the patient back on track.

Ref: Adapted from Kay J & Tasman A. Essentials of Psychiatry, 2nd edition, 2006. John Wiley & Sons, Ltd.

The correct answer is: You seem very upset when talking about your girlfriend. Are you having any problems with your relationship? → Interpretation, The psychiatrist uses approval nods, leaning forwards slightly to express interest, using responses such as 'Yes. And then?' 'Yeah, go on..', 'Uh-huh', etc. → Facilitation, You mentioned having taken overdoses since your last admission. How many times have you taken overdoses in total? Seven...? Eight...? → Symptom exaggeration, What sort of recreational drugs do you usually use when you are drinking? → Symptom expectation, The patient said 'It is not good if one's parents are divorced even before one goes to school.' The doctor replied 'I'd like to hear more about your parents, but first let me get a picture of what's happening to you of late.' → Redirection

Question 7

Not answered

Marked out of 4.00

Varieties of Delusions

For each of the descriptions below choose the most closely associated psychopathological terms from the given list.

A 32-year-old psychotic patient with persistent auditory hallucinations is now developing a belief that a transmitter has been placed in his ear.	Delusional elaboration
A 21-year-old psychotic patient remembers his parents taking him for an operation as a child becoming convinced that he had been implanted with monitoring devices, which have become active in his life.	Delusional memory
A 29-year-old psychotic patient with an odd belief that certain individuals are not who they externally appear to be.	Delusional misidentification
A 33-year-old psychotic patient is convinced that the Mafia is after him and are conspiring to ruin his life	Persecutory delusions

Delusion of reference
Delusions of grandiosity
delusional perception
Delusions of infestations
Autochthonous delusions
Delusions of control

Your answer is incorrect.

Explanation:

Delusional elaboration refers to secondary delusions that arise in a manner which is understandable as the patient attempting to find explanations for primary psychopathological processes (e.g. a patient with persistent auditory hallucinations developing a belief that a transmitter has been placed in their ear).

A delusional memory is a primary delusion that is recalled as arising as a result of a memory (e.g. a patient who remembers his parents taking him to hospital for an operation as a child becoming convinced that he had been implanted with control and monitoring devices that have become active in his adult life).

Delusional misidentification is a delusional belief that certain individuals are not who they externally appear to be. Types of delusional misidentification syndromes include Capgras, Fregoli, Intermetamorphosis, Subjective doubles, Autoscopic, Reverse subjective double, and Reverse Fregoli.

A persecutory delusion is a delusional belief that one's life is being interfered with in a harmful way.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 106, 115, 240.

The correct answer is: A 32-year-old psychotic patient with persistent auditory hallucinations is now developing a belief that a transmitter has been placed in his ear. → Delusional elaboration, A 21-year-old psychotic patient remembers his parents taking him for an operation as a child becoming convinced that he had been implanted with monitoring devices, which have become active in his life. → Delusional memory, A 29-year-old psychotic patient with an odd belief that certain individuals are not who they externally appear to be. → Delusional misidentification, A 33-year-old psychotic patient is convinced that the Mafia is after him and are conspiring to ruin his life → Persecutory delusions

Question 8

Not answered

Marked out of 1.00

Which of the following subtests of Wechsler Adult Intelligence Scale (WAIS) is noted to resist any decline with aging?

Select one:

- Digit symbol
- Block design
- Digit span
- Similarities
- Picture completion

Your answer is incorrect.

Explanation:

Hold tests are supposed to be resistant to age-related decline; so they may be sensitive to organic brain damage such as dementia. The current version of Wechsler Adult Intelligence Scale (WAIS-IV) yields an overall IQ score, plus four index scores and scores on component subtests including: Verbal Comprehension Scale (similarities, vocabulary, information, comprehension); Perceptual Reasoning Scale (block design, matrix reasoning, visual puzzles, picture completion, figure weights); Working Memory Scale (digit span, arithmetic, letter-number sequencing); Processing Speed Scale (symbol search, coding, cancellation).

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Picture completion

Question 9

Not answered

Marked out of 1.00

Hayling test in neuropsychological assessments is used to test which of the following?

Select one:

- Digit span and executive function
- Verbal fluency and performance IQ
- Semantic inhibition and executive function
- Premorbid IQ
- Verbal memory

Your answer is incorrect.

Explanation:

Failures in semantic inhibition are frequently measured with the Hayling Sentence Completion test. In the Hayling test, the individual is presented with 30 sentences where the last word is missing. In the first 15 sentences (Automatic section), the word is strongly cued by the preceding context, and the individual must complete the sentence by producing the missing word. In the next 15 sentences (Inhibition section), a word that makes no sense in the sentence context or is unrelated to the target word must be given by the individual. Thus, in order to complete the sentence, the individual has to inhibit the automatic response and generate a new unrelated word.

Ref: Cervera-Crespo T, González-Alvarez J. Age and semantic inhibition measured by the hayling task: a meta-analysis. Archives of Clinical Neuropsychology. 2017 Mar 1;32(2):198-214.

The correct answer is: Semantic inhibition and executive function

Question 10

Not answered

Marked out of 1.00

A 49-year-old patient with schizophrenia scored poorly on Brixton test. This shows deficits in

Select one:

- Attention and concentration
- Verbal memory
- Visuospatial sequencing
- Abstraction
- Language

Your answer is incorrect.

Explanation:

The Brixton Spatial Anticipation Test is a well-established test of executive function that evaluates the capacity to abstract, follow, and switch rules. It bears similarities to the Wisconsin Card Sorting Test, in that it evaluates the ability to acquire and follow rules, to switch between rules, and to inhibit earlier response patterns. It makes minimal verbal demands. The test consists of 10 numbered circles, one of which is coloured blue. The participant is required to anticipate the position of the blue circle on each successive page of a 55-page test booklet by identifying patterns that vary over the course of the task. Thus, it is a visuospatial sequencing task that measures the ability to learn rules from sequences of stimuli. It is often used to assess executive functioning in people with neurological disorders such as strokes, dementia, and psychiatric disorders such as schizophrenia.

Ref: Rao SP et al. Distinct performance profiles on the Brixton test in frontotemporal dementia. Journal of Neuropsychology. 2021 Jun;15(2):162-85.

The correct answer is: Visuospatial sequencing

Question 11

Not answered

Marked out of 1.00

During extended cognitive assessment by a neuropsychologist, the subject is asked to read a short story containing 25 elements and the psychologist tested both immediate and delayed recall after 30 minutes. The test done by the neuropsychologist is called

Select one:

- Halstead - Reitan scale
- Rivermead behavioural memory test
- Wechsler memory scale
- Wechsler adult intelligence scale
- Rey-Osterrieth test

Your answer is incorrect.

Here the subject is asked to read a short story from the Wechsler Memory scale containing 25 elements and both immediate and delayed recall after an interval of 30 minutes is tested. This is one of the verbal subtests used in Wechsler memory scale. WMS-3 consists of tests grouped under two domains; verbal and nonverbal, all of which have immediate and delayed recall and or recognition components. It also has a third group of tests involve .mental control subtests'.

The correct answer is: Wechsler memory scale

Question 12

Not answered

Marked out of 1.00

A psychologist is performing neuropsychological testing in a 45 year old man. Which one of the following is a test of set shifting?

Select one:

- Wechsler memory scale
- Ravens progressive matrices
- Boston naming test
- Wisconsin Card sorting test
- Clock drawing task

Your answer is incorrect.

Explanation:

The Wisconsin card sorting task is a test of frontal lobe functioning. The patient has to determine the rule for card allocation and allocate cards accordingly. When the rule changes, a patient with frontal lobe dysfunction is likely to make more errors (tests response inhibition and set shifting).

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 88.

The correct answer is: Wisconsin Card sorting test

Question 13

Not answered

Marked out of 3.00

Familial risk rates

Choose the most appropriate risk in each of the following situations:

The risk of inheritance in a 30-year-old man whose mother and maternal grandfather developed Alzheimer's disease in their 50s

A 25-year-old woman with a history of bipolar disorder. What is the chance that one of her children may develop a bipolar spectrum disorder?

A 36-year-old woman with unipolar depression. What is the chance that her daughter inherits the same illness?

40-49%
1-5%
70-79%
10-19%
50-59%
5-10%
80-89%
60-69%
30-39%
20-29%

Your answer is incorrect.

For the common late-onset form of Alzheimer's, the risk for those with no first degree relatives (parents or siblings) with a diagnosis of Alzheimer's disease is 7% or less (given normal life expectancy). The risk for those with one first-degree family member is about 15-20%. For second degree relatives (e.g., grandparents) the risk may be about 10%. For monozygotic twins, the risk is 80-85%. But in early-onset cases (<65) strong autosomal dominant tendency is noted increasing the risk to >50%.

Compared with the offspring of control parents, children of parents with bipolar disorder had an increased risk of having a bipolar spectrum disorder (Odds Ratio 13.4). A meta-analysis and more recent studies have reported rates of BP between 4% and 15% (B is the closest answer) in the offspring of parents with BP and between 0% and 2% in the offspring of healthy parents.(Lapalme et al. Can J Psychiatry. 1997;42(6):623-631). Children in families where both parents had bipolar disorders also were more likely than those in families containing one parent with bipolar disorder to develop the condition (Odds Ratio: 3.6) [Birmaher et al. Arch Gen Psychiatry. 2009;66(3):287-296. Children of parents with bipolar disorder have an increased risk of unipolar (11.4%) depression too. Children of parents with unipolar depression have a 5.9 to 18.4 % (median 9.1%: the closest answer is 10-19%) risk of unipolar depression and no increased risk of bipolar disorder.

The correct answer is: The risk of inheritance in a 30-year-old man whose mother and maternal grandfather developed Alzheimer's disease in their 50s → 50-59%, A 25-year-old woman with a history of bipolar disorder. What is the chance that one of her children may develop a bipolar spectrum disorder? → 5-10%, A 36-year-old woman with unipolar depression. What is the chance that her daughter inherits the same illness? → 10-19%

Question 14

Not answered

Marked out of 3.00

Neuroimaging findings

For each of the condition given below choose ONE appropriate finding from the above list

Geriatric depression	Choose...
Variant CJD	Choose...
Auditory hallucinations in schizophrenia	Choose...

Diffuse oedema of cerebral cortex (MRI)
Medical temporal atrophy (MRI)
Bilateral high pulvinar signals (MRI)
Deep white matter hyperintensities (MRI)
Periventricular white matter changes (MRI)
Primary auditory cortex activation (fMRI)
Focal frontal atrophy (MRI)
Focal atrophy of occipital lobes
Increase in the volume of the brain
Impaired dorso-lateral prefrontal function (FMRI)

Your answer is incorrect.

In geriatric depression the common MRI findings include atrophy, ventricular enlargement, lesions in basal ganglia and deep white matter hyper intensities (WMH). WMH occur particularly in the deep subcortical white matter and to a lesser extent in the basal ganglia and periventricular white matter (Harrison, 2002). In major depression, WMH are linked to risk factors for vascular disease, and thus are often seen in the elderly subjects.

In variant CJD, MRI is the most useful supportive diagnostic test. A characteristic abnormality seen in the posterior thalamic region (pulvinar sign) is highly sensitive and specific for variant CJD. The pulvinar sign has been found in more than 90% of pathologically proven vCJD cases. FLAIR sequences of MRI are most likely to show the abnormality.

In patients with auditory hallucinations in Schizophrenia, primary auditory cortex activation is seen in the f-MRI scan (Dierks et al-1999). Neuroimaging studies have shown that auditory hallucinations in schizophrenic patients were associated with increased metabolic activity in the left primary auditory cortex and the right middle temporal gyrus.

Ref: Harrison, PJ (2002). The neuropathology of primary mood disorder. Brain, http://brain.oxfordjournals.org/content/125/7/1428

The correct answer is: Geriatric depression → Deep white matter hyperintensities (MRI), Variant CJD → Bilateral high pulvinar signals (MRI), Auditory hallucinations in schizophrenia → Primary auditory cortex activation (fMRI)

Question 15

Not answered

Marked out of 3.00

Derivatives of neural tube

For each of the following organs choose the neural tube vesicle of origin from the list provided:

Thalamus	Choose...
Hippocampus	Choose...
Cerebellum	Choose...

Metencephalon
Myelencephalon
Telencephalon
Mesencephalon
Diencephalon

Your answer is incorrect.

Explanation:

At 4 weeks, the human brain is divided longitudinally into the prosencephalon (forebrain), mesencephalon (midbrain), and rhombencephalon (hindbrain). These three subdivisions or 'vesicles' divide further into five divisions by 5 weeks, consisting of the prosencephalon, which forms the telencephalon (including cortex, hippocampus, and basal ganglia) and diencephalon (thalamus, subthalamus, hypothalamus), the mesencephalon (midbrain, including the red nucleus), and the rhombencephalon, yielding metencephalon (pons and cerebellum) and myelencephalon (medulla).

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 939.

The correct answer is: Thalamus → Diencephalon, Hippocampus → Telencephalon, Cerebellum → Metencephalon

Question 16

Not answered

Marked out of 3.00

Choosing psychiatric medications

For each of the following descriptions choose the most appropriate medication from the given list

A 24-year-old woman has had two episodes of irritable mood, increased energy, decreased need for sleep, and pressured speech. She was recently started on medication by her psychiatrist to control these symptoms. She presents to her GP now complaining of a significant worsening of her acne since starting this new medication. What drug was she started on?

A 24-year-old man has multiple problems at his workplace due to 'extreme tiredness' during office hours. For several years, he has had severe daytime sleepiness, episodes of falling asleep without warning, and hearing his name being called out loud on many occasions when no one is around. Which of the following treatments may be of benefit to this patient?

Which of the following drugs carried a risk of thromboembolism especially during the first six months of treatment?

Agomelatine
Paroxetine
Reboxetine
Clozapine
Lithium
Fluoxetine
Clonidine
Citalopram
Modafinil
Escitalopram

Choose...

Your answer is incorrect.

Case 1: Several cutaneous side effects are possible with lithium including acne, follicular and maculopapular eruptions (rashes) and alopecia. With the exception of lithium, the other choices do not worsen acne.

Case 2: The description in the questions fits with a diagnosis of narcolepsy. Stimulants such as methylphenidate can reduce daytime sleepiness. Medications that reduce REM sleep, such as TCAs and SSRIs, are used if cataplexy is also present. Modafinil can reduce the number of sleep attacks and can improve psychomotor performance. Modafinil has been approved as a first-line treatment for excessive daytime sleepiness in patients with narcolepsy. Although modafinil can block several monoaminergic transporters, the mechanism underlying the wake-enhancing function of this drug is probably mainly due to dopamine reuptake inhibition, through blockage of the dopamine transporter.

Case 3: A possible association between clozapine and thromboembolism has been suggested. Thromboembolism may be related to clozapine's observed effects on antiphospholipid antibodies and platelet aggregation. It seems most likely to occur in the first 6 months of treatment but can occur at any time.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 416.

Kornum BR et al. Narcolepsy. Nature reviews Disease primers. 2017 Feb 9;3(1):1-9.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 222.

The correct answer is: A 24-year-old woman has had two episodes of irritable mood, increased energy, decreased need for sleep, and pressured speech. She was recently started on medication by her psychiatrist to control these symptoms. She presents to her GP now complaining of a significant worsening of her acne since starting this new medication. What drug was she started on? → Lithium, A 24-year-old man has multiple problems at his workplace due to 'extreme tiredness' during office hours. For several years, he has had severe daytime sleepiness, episodes of falling asleep without warning, and hearing his name being called out loud on many occasions when no one is around. Which of the following treatments may be of benefit to this patient? → Modafinil, Which of the following drugs carried a risk of thromboembolism especially during the first six months of treatment? → Clozapine

Question 17

Not answered

Marked out of 3.00

Partial agonism

Identify the predominant receptor level of action for the following partial agonistic drugs.

Buspirone	Choose...
Clobazam	Choose...
Buprenorphine	Choose...

5HT3 receptor
D-5 receptor
Benzodiazepine receptor
D-1 receptor
kappa opiod receptor
5 HT1A receptor
Omega opioid receptor
5HT2 receptor
D-2 receptor

Your answer is incorrect.

Partial agonists: Aripiprazole - D2; Buspirone - 5HT1A; Clobazam - BDZ receptor; Buprenorphine - opioid receptor mu.

Aripiprazole carries a low risk of extrapyramidal symptoms despite high levels of D2 occupancy (> 90%) at therapeutic doses; this is probably due to its partial-agonist activity at D2 receptors. It is also a partial agonist at 5-HT1A receptors, a high affinity antagonist at 5-HT2A receptors, and a low-/moderate affinity antagonist at H1 and alpha-1 receptors. It has no anticholinergic effect.

Buspirone is classified as an azapirone (or azaspirone) and is chemically distinct from other psychotropic agents. It is a 5-HT1A partial agonist.

Benzodiazepines act via a specific site called omega site in GABA-A complex. All are agonists except some metabolites of clonazepam and clobazam (used for Lennoux-Gastaut syndrome), which are partial agonists.

Buprenorphine is a partial agonist at mu-opioid receptors.

Ref: Zhang L, Wang J, Wang C. Efficacy and safety of antiseizure medication for Lennox–Gastaut syndrome: a systematic review and network meta-analysis. Developmental Medicine & Child Neurology. 2022 Mar;64(3):305-13.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 672.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 176, 478.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 212.

The correct answer is: Buspirone → 5 HT1A receptor, Clobazam → Benzodiazepine receptor, Buprenorphine → Mu opioid receptor

Question 18

Not answered

Marked out of 5.00

Substance misuse- signs & symptoms

Identify the type of illicit substance misuse associated with each of the following clinical signs & symptoms

Nasal mucosal atrophy and septal perforation	Choose...
Vivid afterimages & recurring intense sensations	Choose...
Apathy and amotivational syndrome	Choose...
Water intoxication	Choose...
Pinpoint pupils	Choose...

Heroin
Cannabis
LSD
Amphetamine
Crack cocaine
Ketamine
Benzodiazepines
Ecstasy

Your answer is incorrect.

Explanation:

Cocaine snorting produces nasal mucosal atrophy and nasal perforation.

Long after ingesting a hallucinogen like LSD, a person can experience a recurrence of hallucinogenic symptoms. This condition can be diagnosed as hallucinogen persisting perception disorder using DSM-5. This reexperiencing of perceptual symptoms following cessation of hallucinogen use is characterized by seeing geometric hallucinations, false perceptions of movement in the peripheral visual fields, flashes or intensified colours, trails of images of moving objects, positive afterimages, halos around objects, macropsia, and micropsia. This is also called flashback phenomenon.

Chronic cannabis use is can cause lethargy, apathy, loss of interest, anergia, reduced drive and lack of ambition, which is collectively known as the amotivational syndrome. Some authors do not agree with the validity of this syndrome.

Water intoxication associated with drug abuse is typically caused by MDMA ("ecstasy"), often secondary to excessive sweating. MDMA is commonly used as a party drug; so abusers often dance for several hours, after preparing themselves by drinking a lot of water beforehand. As a lot of salts is lost through the sweat but does not get replenished by water loading, the osmotic balance is affected, resulting in water

intoxication.

Acute opioid intoxication (e.g. through the use of heroin) is characterized by bradycardia, hypotension, respiratory depression, subnormal body temperature and pin-point pupils.
[Ref: Gowing LR et al. (2002) The health effects of ecstasy: a literature review. Drug Alcohol Rev 21: 53-63]

The correct answer is: Nasal mucosal atrophy and septal perforation → Crack cocaine, Vivid afterimages & recurring intense sensations → LSD, Apathy and amotivational syndrome → Cannabis, Water intoxication → Ecstasy, Pinpoint pupils → Heroin

Question 19

Not answered

Marked out of 4.00

Culture bound syndromes - Descriptions

What is the most appropriate diagnosis in the following cases?

A 25-year-old Srilankan man presenting with lethargy, sleep disturbance and impotence which he attributes to the loss of semen

A 70-year-old Malaysian woman presenting with dissociation, automatic obedience and echolalia in response to the news of her son's death

A 25-year-old Japanese woman who has a persistent and overwhelming fear that her appearance and bodily odour is offensive to other people

A 50-year-old Ethiopian man has an experience of spirit possession. His symptoms include episodes of laughing inappropriately, shouting and hitting his head against a wall

Amok
Dhat
Zar
Taijin Kyofusho
Susto
Frigophobia
Windigo
Latah
Brain Fag
Koro

Your answer is incorrect.

Explanation:

Dhat/Jiryan/Sukra-Prameha refers to semen-loss syndrome noted in the Indian subcontinent and in South Asia. It is rooted in Ayurvedic traditions about bodily production of semen as representing an end-product of energy-demanding metabolism. Dhat is recognised by a whitish discharge in the urine assumed to be semen. This 'loss of semen' is associated with somatic symptoms (weakness, exhaustion), severe anxiety, hypochondriasis, and sexual dysfunction.

Latah has been found mainly in Malaysian and Indonesian cultures in which a person - after experiencing a sudden fright or shock - displays abnormal suggestibility accompanied by echolalia, echopraxia, autonomic obedience, and a state resembling a trance. Originally reported as a common phenomenon among young women in Malaysia, it is becoming less prevalent, (particularly in urban areas) and is now found almost exclusively among older women who have known of the phenomenon since they were young.

Taijin-kyofu(shu), an Asian culture-related psychiatric condition, is a form of social anxiety, defined as a persistent and excessive fear of causing offense to others because of physical characteristics (such as blushing, defect in appearance, staring inappropriately, or emitting foul odours).

Zar is an exclusively North African syndrome: where subjects present with an experience of spirit possession. In Ethiopia and other areas of North Africa, it is characterised by episodes of dissociation that are attributed to spirit possession and are associated with signs and symptoms such as shouting, laughing, self-injury or self-mutilation, singing, and weeping, often followed by apathy and withdrawal.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 990.

Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

Tseng WS. From peculiar psychiatric disorders through culture-bound syndromes to culture-related specific syndromes. Transcultural psychiatry. 2006 Dec;43(4):554-76.

Skimming KA, Miller CW. Transdiagnostic approach to olfactory reference syndrome: neurobiological considerations. Harvard review of psychiatry. 2019 May 1;27(3):193-200.

The correct answer is: A 25-year-old Srilankan man presenting with lethargy, sleep disturbance and impotence which he attributes to the loss of semen → Dhat, A 70-year-old Malaysian woman presenting with dissociation, automatic obedience and echolalia in response to the news of her son's death → Latah, A 25-year-old Japanese woman who has a persistent and overwhelming fear that her appearance and bodily odour is offensive to other people → Taijin Kyofusho, A 50-year-old Ethiopian man has an experience of spirit possession. His symptoms include episodes of laughing inappropriately, shouting and hitting his head against a wall → Zar

Question 20

Not answered

Marked out of 1.00

Transcriptome is a

Select one:

- Statistical test
- Type of plasmid
- Collection of expressed m-RNA
- Positional cloning
- Stretch of DNA

Your answer is incorrect.

A transcriptome is different from a genome, the latter being the entire DNA sequence of an organism. A transcriptome represents the very small percentage of the genome - less than 5 percent in humans - that is transcribed into RNA molecules. A gene may produce many different types of mRNA molecules, so a transcriptome is much more complex than the genome that encodes it.

The correct answer is: Collection of expressed m-RNA

Question 21

Not answered

Marked out of 1.00

A nonsense mutation refers to

Select one:

- A point mutation
- Substitution of stop codons
- Destruction of stop codons
- Deletion of stop codons
- Creation of stop codons

Your answer is incorrect.

Explanation:

A nonsense mutation is any mutation that produces one of the stop codons, causing premature termination of protein synthesis by the gene in which it occurs.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Creation of stop codons

Question 22

Not answered

Marked out of 1.00

The gene most strongly implicated in ADHD is

Select one:

- Dopamine 2 receptor gene
- Dopamine 3 receptor gene
- Dopamine 4 receptor gene
- Dopamine 1 receptor gene
- Dopamine 5 receptor gene

Your answer is incorrect.

Explanation:

Family studies and population-based studies have found an association between the dopamine 4 receptor seven-repeat allele (DRD4) gene and ADHD.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 139.

The correct answer is: Dopamine 4 receptor gene

Question 23

Not answered

Marked out of 1.00

The structure of the brain involved in taste sensation is

Select one:

- Amygdala
- Hippocampus
- Prefrontal cortex
- Entorhinal cortex
- Anterior insula

Your answer is incorrect.

Explanation:

Composed of three short gyri and an additional ventral accessory gyrus, the anterior insula is home to the gustatory complex, which is integral to the representation of tastes, oral texture, and temperature.

Ref: Yiannakas A, Rosenblum K. The insula and taste learning. Frontiers in molecular neuroscience. 2017 Nov 3;10:335.

The correct answer is: Anterior insula

Question 24

Not answered

Marked out of 1.00

Which of the following areas of the brain does not lack blood brain barrier?

Select one:

- Median eminence
- Posterior pituitary
- Area postrema
- Subfornical organ
- Anterior pituitary

Your answer is incorrect.

The anterior pituitary does not lack BBB. There are some areas of the brain that do not have a blood- brain barrier. These are so-called circum-ventricular organs e.g. subfornical organ, area postrema (chemo receptor trigger zone), median eminence and posterior pituitary. Here the capillaries are fenestrated like those in peripheral tissues

The correct answer is: Anterior pituitary

Question 25

Not answered

Marked out of 1.00

The white matter tract connecting Wernicke's are and Broca's area is called

Select one:

- Cingulate gyrus
- Anterior commissure
- Arcuate fasciculus
- Habenular commissure
- Planum temporale

Your answer is incorrect.

Explanation:

The arcuate fasciculus is a small bundle of nerve fibres in each cerebral hemisphere that curves round the back of the lateral sulcus and that in the (usually) left hemisphere connects Broca's area with Wernicke's area. Lesions to it often result in conduction aphasia.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Arcuate fasciculus

Question 26

Not answered

Marked out of 1.00

Which hypothalamic nucleus regulates prolactin secretion?

Select one:

- Medial dorsal nucleus
- Suprachiasmatic nucleus
- Preoptic nucleus
- Subthalamic nucleus
- Arcuate nucleus

Your answer is incorrect.

Explanation:

The arcuate nucleus is a small nucleus near the base of the hypothalamus containing the cell bodies that produce hypothalamic hormones. The prolactin-releasing factor is found in the arcuate nucleus. This activates the lactotrophic cells of the anterior pituitary. Dopamine, via the tuberoinfundibular dopaminergic system, regulates prolactin secretion through direct projection to the pituitary.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Arcuate nucleus

Question 27

Not answered

Marked out of 1.00

Which of the following cells help in regulating potassium ion concentration in the extracellular space around neuronal cell body?

Select one:

- Purkinje cells
- Astrocytes
- Microglia
- Schwann cells
- Oligodendrocytes

Your answer is incorrect.

Explanation:

Several studies underline that reactive astrocytes alter homeostatic functions such as potassium ion uptake, ion buffering, calcium signalling, and excitatory neurotransmitter uptake.

Ref: Siracusa R et al. Astrocytes: role and functions in brain pathologies. Frontiers in pharmacology. 2019 Sep 27;10:1114.

The correct answer is: Astrocytes

Question 28

Not answered

Marked out of 1.00

Acetylcholine is a substrate of which of the following substances?

Select one:

- Choline Acetyl transferase
- Acetyl cholinesterase
- Monoamine oxidase
- Acetyl Co A
- Choline

Your answer is incorrect.

Explanation:

Acetylcholine is formed in cholinergic neurons from two precursors: choline and acetyl coenzyme A (AcCoA). Choline is derived from dietary and intraneuronal sources, and AcCoA is made from glucose in the mitochondria of the neuron. These two substrates interact with the synthetic enzyme choline acetyltransferase (ChAT) to produce the neurotransmitter acetylcholine (ACh). ACh's actions are terminated by one of two enzymes, acetylcholinesterase or butyrylcholinesterase. Both enzymes convert ACh into choline, which is then transported back into the presynaptic cholinergic neuron for resynthesis into ACh.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 505.

The correct answer is: Acetyl cholinesterase

Question 29

Not answered

Marked out of 1.00

Which enzyme is required for the metabolism of the transmitter at the neuromuscular junction?

Select one:

- Choline acetyltransferase
- Alpha ketoglutarate
- Glutamine synthetase
- Acetylcholinesterase
- Glutaminase

Your answer is incorrect.

Explanation:

Acetylcholinesterase is an essential enzymatic component of the neuromuscular junction where it is responsible for terminating neurotransmission by the cholinergic motor neurons. The enzyme at the neuromuscular junction is contributed primarily by the skeletal muscle where is it produced at higher levels in the post-synaptic region of the fibres.

Ref: Rotundo RL. The NMJ as a model synapse: New perspectives on formation, synaptic transmission and maintenance: Acetylcholinesterase at the neuromuscular junction. Neurosci Lett. 2020 Sep 14;735:135157. doi: 10.1016/j.neulet.2020.135157. Epub 2020 Jun 12. PMID: 32540360.

The correct answer is: Acetylcholinesterase

Question 30

Not answered

Marked out of 1.00

The rate limiting enzyme in alcohol metabolism is

Select one:

- Acetaldehyde dehydrogenase
- Alanine dehydrogenase
- DOPA dehydrogenase
- Alcohol dehydrogenase
- Aldehyde dehydrogenase

Your answer is incorrect.

The first step in the metabolism of alcohol is the oxidation of ethanol to acetaldehyde catalyzed by alcohol/dehydrogenase containing the coenzyme NAD+. The acetaldehyde is further oxidized to acetic acid and finally CO2 and water through the citric acid cycle. A number of metabolic effects from alcohol are directly linked to the production of an excess of both NADH and acetaldehyde

The correct answer is: Alcohol dehydrogenase

Question 31

Not answered

Marked out of 1.00

Which among the following statements favour the dopamine hypothesis of schizophrenia?

Select one:

- Amphetamines release Dopamine into the synapse and cause positive schizophrenic symptoms
- Disulfiram stimulates Dopamine metabolism and may exacerbate schizophrenia
- Postal Emission Tomography (PET) shows decreased Dopamine receptor binding in drug naïve patients.
- Phencyclidine (PCP) causes negative schizophrenic symptoms
- Both alpha and beta isomers of Flupenthixol are effective anti psychotic drugs

Your answer is incorrect.

Explanation:

The simplest formulation of the dopamine hypothesis of schizophrenia posits that schizophrenia results from too much dopaminergic activity. The theory evolved from two observations. First, the efficacy and the potency of many antipsychotic drugs correlate with their ability to act as an antagonists of the dopamine type 2 receptor. Second, drugs that increase dopaminergic activity, notably cocaine and amphetamine, are psychotomimetic.

Only the alpha isomer of Flupentixol is a Dopamine antagonist and an effective anti psychotic drug. Phencyclidine causes both positive and negative symptoms of schizophrenia but via antagonisms at the NMDA receptor. Disulfiram actually inhibits Dopamine metabolism and may exacerbate schizophrenia. PET studies show increased Dopamine receptor binding in drug-naive patients.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 361.

The correct answer is: Amphetamines release Dopamine into the synapse and cause positive schizophrenic symptoms

Question 32

Not answered

Marked out of 1.00

Release of transmitters synthesized by neurons in the central nervous system is directly regulated by the concentration of which of the following electrolytes?

Select one:

- Calcium influx
- Chloride influx
- Potassium influx
- Magnesium influx
- Sodium influx

Your answer is incorrect.

Action potentials at the presynaptic axon terminals open up calcium channels, permitting calcium influx. This leads to movement of synaptic vesicles and subsequent release of stored neurotransmitters.

The correct answer is: Calcium influx

Question 33

Not answered

Marked out of 1.00

The rate limiting step in the synthesis of serotonin is mediated by which of the following enzymes?

Select one:

- L-tryptophan hydroxylase
- Monoamine oxidase A
- Tyrosine Hydroxylase
- Aminoacid decarboxylase
- Acetyl CO-A Synthetase

Your answer is incorrect.

Explanation:

The rate-limiting step in serotonin synthesis is the hydroxylation of tryptophan by the enzyme tryptophan hydroxylase to form 5-hydroxytryptophan. Two isoforms of tryptophan hydroxylase exist – one isoform is found mainly in the periphery, whereas the second isoform is restricted to the CNS.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 957.

The correct answer is: L-tryptophan hydroxylase

Question 34

Not answered

Marked out of 1.00

In schizophrenia, dopamine deficiency is seen in which of the following brain regions?

Select one:

- Mesolimbic pathway
- Mesocortical projections
- Tuberoinfundibular pathway
- Nigrostriatum
- All of the listed options

Your answer is incorrect.

Explanation:

The mesocortical dopamine (DA) pathway arises from cell bodies in the ventral tegmental area, projecting to areas of the prefrontal cortex. Branches of this pathway into the dorsolateral prefrontal cortex are hypothesised to regulate cognition and executive functions, whereas branches of this pathway into the ventromedial parts of the prefrontal cortex are hypothesised to regulate emotions and affect. The exact role of the mesocortical DA pathway in mediating symptoms of schizophrenia is still a matter of debate, but many researchers believe that cognitive and some negative symptoms of schizophrenia may be due to a deficit of DA activity in mesocortical projections to the dorsolateral prefrontal cortex whereas affective and other negative symptoms of schizophrenia may be due to a deficit of DA activity in mesocortical projections to the ventromedial prefrontal cortex.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 95.

The correct answer is: Mesocortical projections

Question 35

Not answered

Marked out of 1.00

Which of the following is an agonist at the mu-opioid receptor?

Select one:

- Disulfiram
- Naltrexone
- Buprenorphine
- Methadone
- Naloxone

Your answer is incorrect.

Explanation:

Methadone is a full agonist at mu opioid receptors and has a long half-life (24 hours). Buprenorphine is a partial opioid agonist.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 478.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 636.

The correct answer is: Methadone

Question 36

Not answered

Marked out of 1.00

A 75-year-old gentleman with memory problems, disinhibition and lack of insight. What microscopic findings will be expected in this case?

Select one:

- Spongiform encephalopathy
- Alpha synuclein deposits
- Intraneuronal inclusions
- Beta amyloid peptides
- Cytoplasmic inclusions

Your answer is incorrect.

This man is exhibiting symptoms of fronto-temporal dementia. The syndrome of FTD has been associated with two types of pathological change, which in individual cases cannot be predicted on clinical grounds. Tau positive inclusions account for the majority of cases, and most familial (but not sporadic) cases with tau pathology carry mutations in the microtubule-associated tau protein on chromosome 17. The other main pathological finding in FTD is so-called ubiquitin-only pathology with cytoplasmic and nuclear inclusions (i.e. negative for tau or alpha-synuclein), suggesting a link with Amyotrophic lateral sclerosis.

The correct answer is: Cytoplasmic inclusions

Question 37

Not answered

Marked out of 1.00

A 37-year-old woman presents initially with optic neuritis and then with episodes of waxing and waning neurological deficits. The most likely pathological change in brain is

Select one:

- Gliosis
- Demyelination
- Lewy bodies
- Neuritic plaques
- Amyloid plaques

Your answer is incorrect.

Explanation:

This patient has Multiple Sclerosis (MS), which is twice as common in women as in men. MS is an immune-mediated demyelinating disease. 85-90% of cases start with an episode of acute neurological symptoms (e.g., optic neuritis). Most patients follow a relapsing-remitting course.

Ref: Ron MA. The neuropsychiatry of multiple sclerosis. BJPsych Advances. 2018 May;24(3):178-87.

The correct answer is: Demyelination

Question 38

Not answered

Marked out of 1.00

A newborn child suffers from seizures and hydrocephalus. On examination, a midline facial anomaly and spasticity are evident. Further investigations reveal agenesis of corpus callosum. Which of the following underlying pathology is most likely?

Select one:

- Tissue loss secondary to obstetric trauma
- Disorder of neuronal migration
- Congential transmission of infection
- Excessive synaptic pruning
- Neurodegeneration

Your answer is incorrect.

Agenesis of the corpus callosum (ACC) has a varied background, ranging from maternal drinking, which is responsible for many defects in neural development, to prenatal infections, chromosomal errors, or gene mutations. The latter can present as isolated ACC or can be combined with other malformations that lead to syndromes (e.g., Coffin-Siris Syndrome). The most frequent causes of ACC are gene mutations that are related to pathways of axon guidance, ciliary development, cell adhesion, proliferation, differentiation, and migration.

Ref: Hofman J et al. Corpus callosum agenesis: an insight into the etiology and spectrum of symptoms. Brain Sciences. 2020 Sep;10(9):625.

http://www.ninds.nih.gov/disorders/neuronal_migration/neuronal_migration.htm

The correct answer is: Disorder of neuronal migration

Question 39

Not answered

Marked out of 1.00

In which of the following conditions brain biopsy can show lesions with intracytoplasmic inclusion bodies?

Select one:

- Alzheimer's disease
- Lewy body dementia
- Huntington's disease
- Tuberculous meningitis
- CJD

Your answer is incorrect.

Explanation:

Pathological features of Lewy body dementia include eosinophilic A-synuclein neuronal inclusions (Lewy bodies), with neuronal loss in brainstem nuclei (especially basal ganglia) and paralimbic and neocortical structures.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 162.

The correct answer is: Lewy body dementia

Question 40

Not answered

Marked out of 1.00

Which of the following EEG wavelets have low amplitude and high frequency?

Select one:

- Sigma waves
- Theta waves
- Beta waves
- Spike waves
- Alpha waves

Your answer is incorrect.

Explanation:

Beta waves have high (more than 13Hz) frequency but low amplitude.

EEG Rhythm	Frequency range	Amplitude	Main scalp area	Subject condition
Delta	0.1-3.5 Hz	50-350 uV	Variable	Drowsiness and sleep Hyperventilation Infancy and childhood
Theta	4-7.5 Hz	10-150 uV	Variable	Drowsiness and sleep Hyperventilation Infancy and childhood Small amount in awake adults
Alpha	8-13 Hz	20-100 uV	Posterior (occipital-parietal)	Relaxed wakefulness with eyes closed
Beta	Above 13 Hz	10-30 uV	Frontal or diffuse	Increase during cognitive efforts as well as during drowsiness and sleep
Mu	7-12 Hz	10-50 uV	Central	Relaxed wakefulness with both eyes open and closed Blocked by movements of the contralateral body parts
Lambda	Sharp transients of 200-300 msec duration	Below 50 uV	Occipital	Visual exploration

Adapted from Standard electroencephalography in clinical psychiatry: A practical handbook. 2011.

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Ref: Boutros NN, Galderisi S, Pogarell O, Riggio S. Standard electroencephalography in clinical psychiatry: A practical handbook. 2011.

The correct answer is: Beta waves

Question 41

Not answered

Marked out of 1.00

Which of the following is a gut hormone released from small intestine in response to fats and proteins and serves to regulate food intake?

Select one:

- Neurokinin
- Serotonin
- Capsaicin
- Substance P
- Cholecystokinin

Your answer is incorrect.

Explanation:

Cholecystokinin is a peptide hormone released by the small intestine that helps to signal satiety during food consumption.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 479.

The correct answer is: Cholecystokinin

Question 42

Not answered

Marked out of 1.00

The suprachiasmatic nucleus plays an important role in which of the following functions?

Select one:

- Attention
- Long term memory
- Circadian rhythms
- Sexual behaviour
- Water and hunger

Your answer is incorrect.

Explanation:

The key neurobiological regulator of the circadian sleep system is the suprachiasmatic nucleus of the hypothalamus.

Ref: Smith PC, Mong JA. Neuroendocrine control of sleep. Neuroendocrine Regulation of Behavior. 2019:353-78.

The correct answer is: Circadian rhythms

Question 43

Not answered

Marked out of 1.00

Miss. X suffers from eating disorder characterized by binge eating, body image disturbances and measures taken to reduce weight . This condition is associated with reduced levels of

Select one:

- Dopamine
- Glycine
- Vasopressin
- Leptin
- Endorphins

Your answer is incorrect.

Explanation:

Changes in leptin and ghrelin occur in patients with anorexia or bulimia nervosa and have been suggested to represent not only homeostatic adaptations to an altered energy balance but to contribute also to the acquisition and/or maintenance of persistent starvation, binge eating, and physical hyperactivity, which are potentially rewarding for eating disordered patients.

Ref: Monteleone AM et al. Neuroendocrinology and brain imaging of reward in eating disorders: a possible key to the treatment of anorexia nervosa and bulimia nervosa. Progress in neuro-psychopharmacology and biological psychiatry. 2018 Jan 3;80:132-42.

The correct answer is: Leptin

Question 44

Not answered

Marked out of 1.00

REM sleep behaviour disorder is associated with

Select one:

- Alzheimer's disease
- Fronto temporal dementia
- Wilson's disease
- Lewy body disease
- Huntington's disease

Your answer is incorrect.

Explanation:

REM sleep behaviour disorder (RBD) is a parasomnia manifested by recurrent dream enactment behaviour associated with an absence of normal REM sleep atonia. It occurs in 76% of cases of dementia with Lewy bodies. It is also associated with other disorders such as Parkinson's disease, multi-system atrophy and Guillain Barre syndrome. RBD may be seen in the prodrome of DLB and Parkinson's disease.

Ref: McKeith IG et al. Diagnosis and management of dementia with Lewy bodies: Fourth consensus report of the DLB Consortium. Neurology. 2017 Jul 4;89(1):88-100.

The correct answer is: Lewy body disease

Question 45

Not answered

Marked out of 1.00

Phenothiazines that are associated with pigment deposits in anterior lens capsule include

Select one:

- Chlorpromazine and promazine
- Perphenazine and trifluoperazine
- Chlorpromazine and perphenazine
- Promazine and thioridazine
- Chlorpromazine and thioridazine

Your answer is incorrect.

Explanation:

High-dose chlorpromazine can result in abnormal pigmentation of the eyelids, cornea, and lens. However, this is not associated with visual disturbance and resolves on cessation of treatment. Thioridazine (now withdrawn) was also associated with pigmentary retinopathy. Corneal oedema can also (rarely) occur with chlorpromazine, leading to loss of vision if treatment is not withdrawn.

Ref: Taylor DM, Gaughran F, Pillinger T. The Maudsley Practice Guidelines for Physical Health Conditions in Psychiatry. 2021. p. 519.

The correct answer is: Chlorpromazine and thioridazine

Question 46

Not answered

Marked out of 1.00

Which of the following is a risk factor for neuroleptic-induced akathisia?

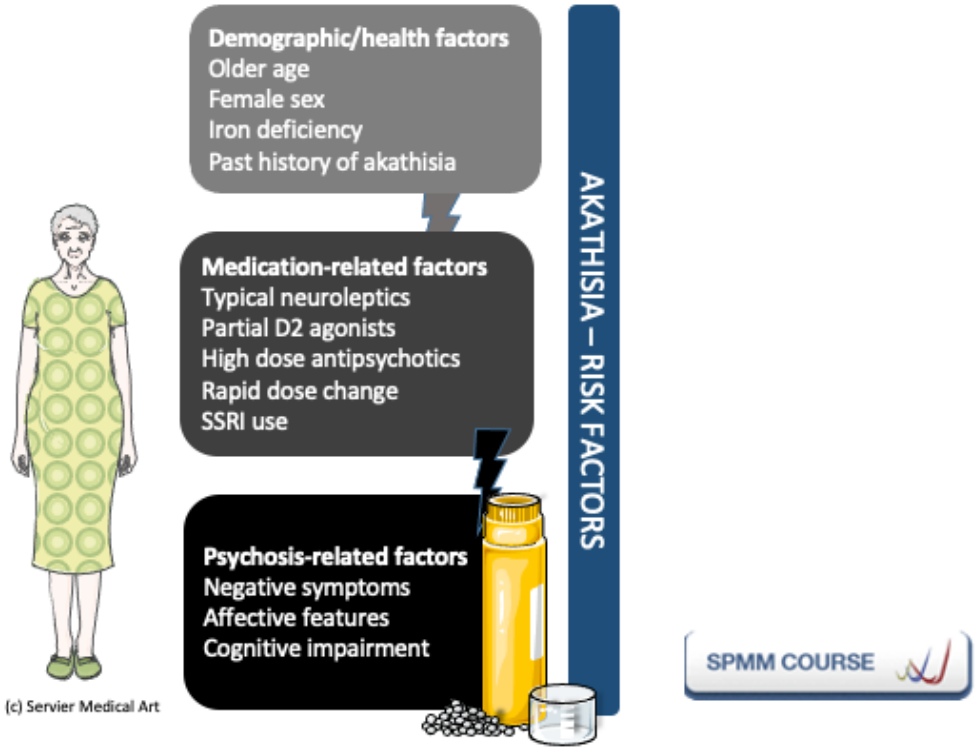
Select one:

- Non-affective psychosis
- Afro-Caribbean ethnicity
- Younger age
- Lipid abnormalities
- Female gender

Your answer is incorrect.

Explanation:

Female gender and older age are risk factors for akathisia.



The correct answer is: Female gender

Question 47

Not answered

Marked out of 1.00

In the setting of tardive dyskinesia, switching or withdrawing antipsychotic medication is not always effective or advisable. Which of the following has been shown to be an effective add-on agent for the treatment of tardive dyskinesia?

Select one:

- St. John's Wort
- Folate
- Pyridoxine (vitamin B6)
- Vitamin B12
- Hormone replacement therapy

Your answer is incorrect.

Explanation:

The use of pyridoxine (vitamin B6) as an add-on agent for the treatment of tardive dyskinesia is supported by Cochrane and meta-analysis data. It can be dosed up to 400mg/day.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 118.

The correct answer is: Pyridoxine (vitamin B6)

Question 48

Not answered

Marked out of 1.00

The most well-established risk factor in terms of developing tardive dyskinesia when using antipsychotics is

Select one:

- Brain pathology
- Mood disorder
- Increasing age
- Afro Caribbean ethnicity
- Female gender

Your answer is incorrect.

Explanation:

The most consistent factor predisposing to the development of tardive dyskinesia is age - the older the patient, the greater the risk. There is up to 6-fold greater incidence in those first exposed to antipsychotics in middle to late life. Other risk factors include antipsychotic potency, number of antipsychotic-free intervals, previous (early) extrapyramidal symptoms, affective disorders, organicity (leucotomy, structural brain changes, negative (deficit) symptoms), and genetics.

Ref: Owens DC. Tardive dyskinesia update: the syndrome. BJPsych Advances. 2019 Jan;25(1):57-69.

The correct answer is: Increasing age

Question 49

Not answered

Marked out of 1.00

Fetal Ebstein anomaly is associated with which of the following psychotropic medications?

Select one:

- Lithium
- Sodium valproate
- Lamotrigine
- Carbamazepine
- Gabapentin

Your answer is incorrect.

Explanation:

Much attention has been focused on the rare but severe cardiac anomaly of the Ebstein type. There may be an increased risk of **Ebstein anomaly** (prevalence in the **general population** of 1:20 000) if lithium taken during 1st trimester (maximum risk 2-6 weeks after conception), though the size of risk uncertain. More recent data suggest the magnitude of effect is **much smaller** than previously estimated. Recent data have shown an **association with maternal mental health problems** generally rather than specifically with lithium.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 690.

McAllister-Williams RH et al. British Association for Psychopharmacology consensus guidance on the use of psychotropic medication preconception, in pregnancy and postpartum 2017. Journal of Psychopharmacology. 2017 May;31(5):519-52.

The correct answer is: Lithium

Question 50

Not answered

Marked out of 1.00

A woman comes into Emergency Department with vomitting, diarrhoea and severe tremors with weight loss on a hot afternoon. She is taking some unknown psychotropic medication for a long time. She is recently started on 'water tablets' for swelling in her legs. Which of the following is a likely cause of her symptoms?

Select one:

- Hypokalemia
- Lithium toxicity
- Hyponatremia
- Neuroleptic malignant syndrome
- Serotonin syndrome

Your answer is incorrect.

Explanation:

The story here is suggestive of lithium overdose or toxicity. Diuretics can reduce the renal clearance of lithium, the magnitude of this effect being greater with thiazide than loop diuretics. Lithium levels usually rise within 10 days of a thiazide diuretic (e.g. bendroflumethiazide, chlortalidone, inadapamide, xipamide) being prescribed. Although there are case reports of lithium toxicity induced by loop diuretics, many patients receive loop diuretics (e.g. furosemide) together with lithium without apparent problems.

Hot weather with dehydration is another important risk factor for precipitating lithium toxicity. The person can present with vomiting, diarrhoea and severe tremors.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 253.

The correct answer is: Lithium toxicity

Question 51

Not answered

Marked out of 1.00

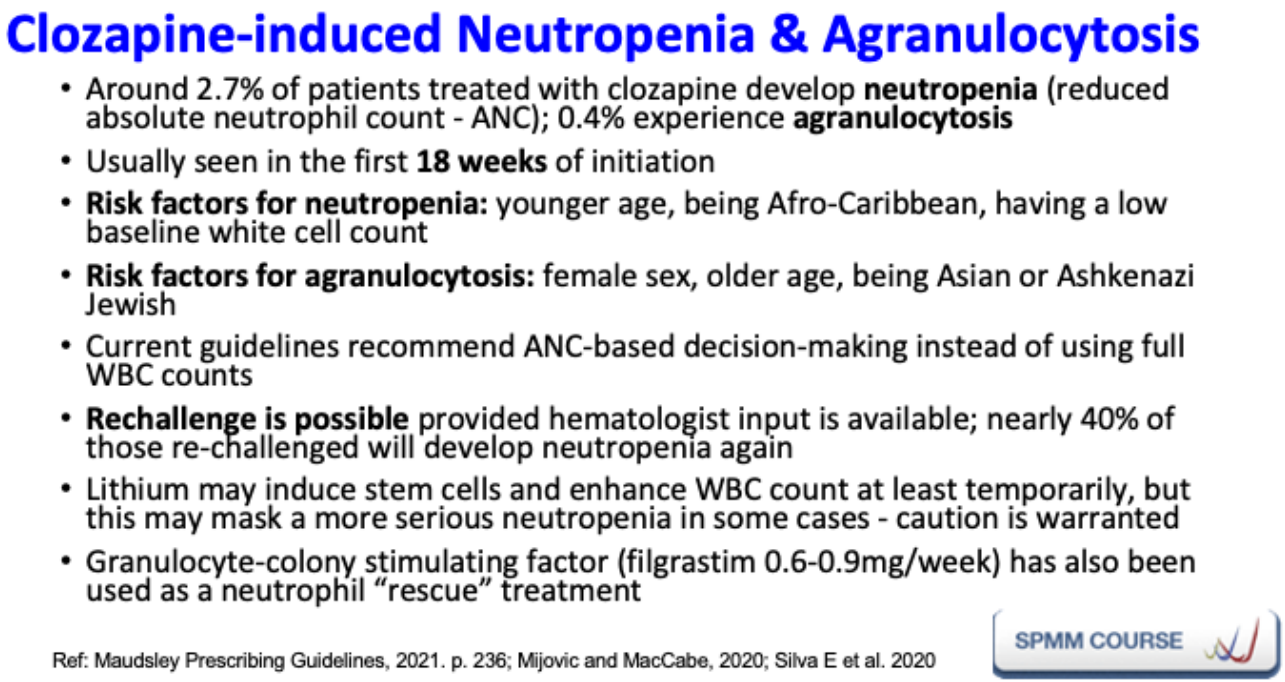
The factors that increase the risk of agranulocytosis in patients treated with clozapine include all of the following except

Select one:

- Female gender
- Male gender
- Increasing age
- Being Ashkenazi Jewish
- Asian ethnicity

Your answer is incorrect.

Explanation:



Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 236.

Mijovic A, MacCabe JH. Clozapine-induced agranulocytosis. Annals of Hematology. 2020 Aug 20:1-6.

Silva E et al. Clozapine rechallenge and initiation despite neutropenia-a practical, step-by-step guide. BMC psychiatry. 2020 Dec;20:1-9.

The correct answer is: Male gender

Question 52

Not answered

Marked out of 1.00

Which one of the following is the least likely feature associated with Neuroleptic malignant syndrome?

Select one:

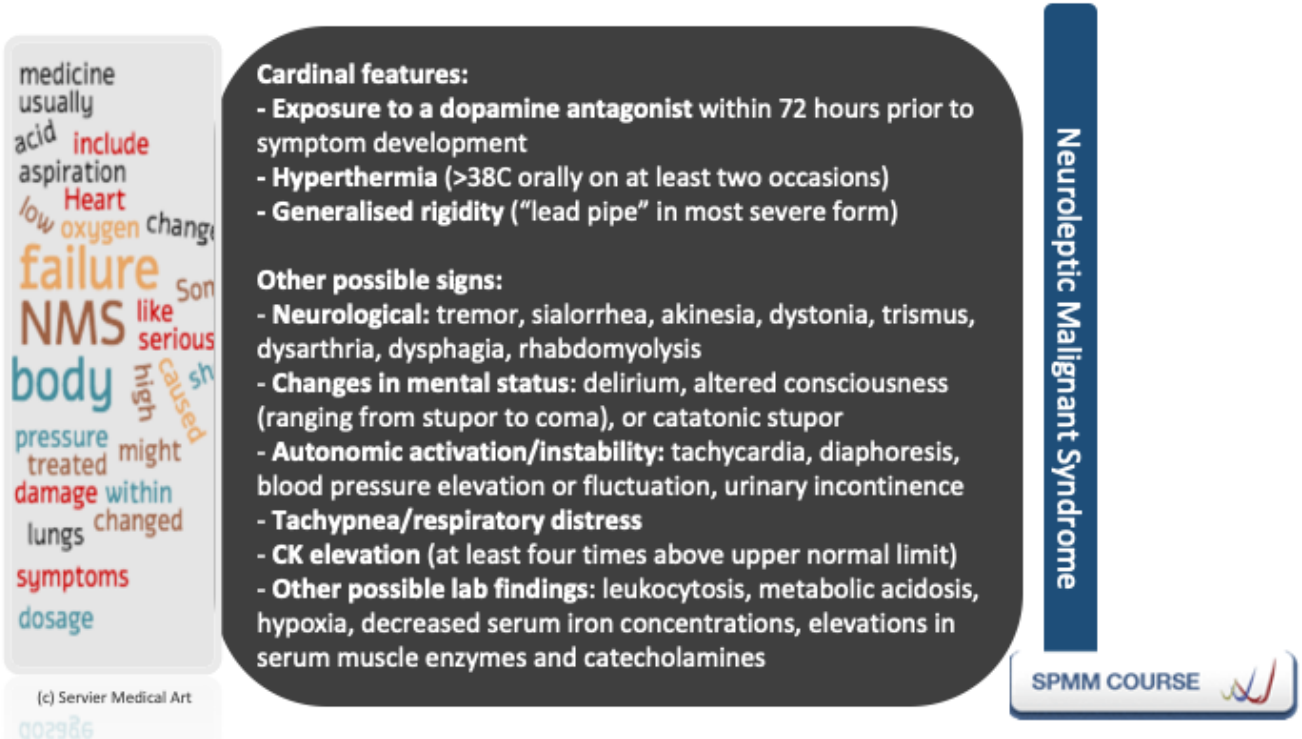
- Muscle rigidity
- Hyperthermia
- Autonomic lability
- Myoclonus
- Raised CPK levels

Your answer is incorrect.

Explanation:

Myoclonus is a feature that could be seen in serotonin syndrome and not in NMS.

Neuroleptic malignant syndrome (NMS):



Differentiating NMS from serotonin syndrome:

Features	NMS	Serotonin Syndrome
Medication History	Dopamine antagonist	Pro-serotonergic medication
Time for Development	1-3 days	<12 hours
Pupils	Normal	Mydriasis (dilated)
Bowel Sounds	Normal or decreased	Hyperactive
Neuromuscular Tone	Rigidity in all muscle groups	Increased - predominantly in lower extremities
Reflexes	Bradyreflexia	Hyperreflexia/clonus (unless masked by inc muscle tone)
Mental Status	Stupor, alert, mutism, coma	Agitation, coma
Laboratory Findings	Increased WBC, high CK	No leukocytosis, CK normal

Ref: Volpi-Abadie et al. 2013, Maudsley Practice Guidelines for Physical Health Conditions in Psychiatry, 2021



Ref: Volpi-Abadie J et al. Serotonin syndrome. Ochsner Journal. 2013 Dec 21;13(4):533-40.

Taylor DM, Gaughran F, Pillinger T. The Maudsley Practice Guidelines for Physical Health Conditions in Psychiatry. 2021.

The correct answer is: Myoclonus

Question 53

Not answered

Marked out of 1.00

A common side effect of sildenafil is

Select one:

- Bradycardia
- Hypotension
- Nasal stuffiness
- Insomnia
- Hair loss

Your answer is incorrect.

Explanation:

Common side effects of phosphodiesterase inhibitors (such as sildenafil and tadalafil) are mild headaches, dizziness, and nasal congestion.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 176.

The correct answer is: Nasal stuffiness

Question 54

Not answered

Marked out of 1.00

Tom recently underwent an alcohol detoxification programme. Upon achieving abstinence, he was keen on anti-craving treatment and has been started on acamprosate. Which one among the following is a common side effect of acamprosate?

Select one:

- Constipation
- Hair loss
- Nausea
- Headache
- Dizziness

Your answer is incorrect.

Explanation:

Acamprosate is relatively well tolerated; adverse effects include diarrhoea, abdominal pain, nausea, vomiting, and pruritus.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 463.

The correct answer is: Nausea

Question 55

Not answered

Marked out of 1.00

Which one among the following has a high hepatotoxic potential?

Select one:

- Trazodone
- Amitriptyline
- Dothiepin
- Imipramine
- Nefazodone

Your answer is incorrect.

Explanation:

Rare cases of sometimes fatal hepatotoxicity led the original manufacturer to discontinue the production of branded nefazodone in 2004. Generic nefazodone remains available in the United States.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 638.

The correct answer is: Nefazodone

Question 56

Not answered

Marked out of 1.00

Low serum iron is associated with which of the following side effects related to neuroleptic medication treatment?

Select one:

- sedation
- orthostatic hypotension
- weight gain
- neuroleptic malignant syndrome
- hyperglycaemia

Your answer is incorrect.

Explanation:

A lesser-known risk factor for neuroleptic malignant syndrome is low serum iron.

Ref: Patil BS et al. Low serum iron as a possible risk factor for neuroleptic malignant syndrome. International Journal of Applied and Basic Medical Research. 2014 Jul;4(2):117.

The correct answer is: neuroleptic malignant syndrome

Question 57

Not answered

Marked out of 1.00

A 45-year-old gentleman with depression has been started on Tranylcypromine. Which drug is safe to be taken?

Select one:

- Nifedipine
- Pethidine
- Pseudoephedrine
- Methadone
- Salbutamol

Your answer is incorrect.

Explanation:

Tranylcypromine is a MAOI Inhibitor. Serotonin toxicity, often referred to as serotonin syndrome, is an iatrogenic drug-induced toxidrome associated with raised intra-synaptic concentrations of serotonin (5-HT) in the CNS. Dangerous toxic reactions, including death, have been shown to result after the administration of meperidine (pethidine – an opioid analgesic drug used in anaesthesia) to patients taking an MAOI such as phenelzine or tranylcypromine. Other opioids to avoid when taking tranylcypromine include tramadol and methadone. Over-the-counter medications to avoid include cold and cough preparations, including those containing dextromethorphan, decongestants (e.g. phenylephrine, pseudoephedrine), hay fever medication, sinus medications, asthma inhalant medications, weight reducing preparations, and 'pep' pills.

Ref: Baldo BA, Rose MA. The anaesthetist, opioid analgesic drugs, and serotonin toxicity: a mechanistic and clinical review. British journal of anaesthesia. 2020 Jan 1;124(1):44-62.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 789.

The correct answer is: Nifedipine

Question 58

Not answered

Marked out of 1.00

According to CATIE study, which drug had the lowest drop out rate when treating schizophrenia?

Select one:

- Amisulpride
- Quetiapine
- Olanzapine
- Risperidone
- Ziprasidone

Your answer is incorrect.

Explanation:

CATIE stands for Clinical Antipsychotic Trials of Intervention Effectiveness. The study design was double-blind pragmatic RCT. 1493 patients with chronic schizophrenia were included and randomised to one of the following: Olanzapine, Quetiapine, Risperidone, Ziprasidone (added later in the trial), Perphenazine. The primary outcome was a 'real-world' measure - discontinuation for any reason (irrespective of whether patient-initiated or physician-initiated). Irrespective of the prescribed drug - 74% discontinued treatment in 18 months (surprisingly high despite naturalistic design). Median time to discontinue was 4.6 months. Olanzapine had lowest discontinuation rate (still 64%) - but highest side effect burden. 64% discontinued olanzapine; 75%, perphenazine; 82%, quetiapine; 74%, risperidone; and 79%, ziprasidone.

Ref: Swartz MS et al. Special section on implications of CATIE: what CATIE found: results from the schizophrenia trial. Psychiatric services. 2008 May;59(5):500-6.

The correct answer is: Olanzapine

Question 59

Not answered

Marked out of 1.00

Which of the following adverse effect of clozapine is mediated via alpha adrenoceptor blockade?

Select one:

- sedation
- weight gain
- constipation
- sialorrhea
- orthostatic hypotension

Your answer is incorrect.

Explanation:

Clozapine's blockade of alpha-1 adrenergic receptors can cause dizziness, hypotension, and syncope.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 193.

The correct answer is: orthostatic hypotension

Question 60

Not answered

Marked out of 1.00

Mr Y is a 45-year-old gentleman with history of schizophrenia. He suffers from akathisia due to taking risperidone and also has some depressive symptoms. The drug of choice in his case would be

Select one:

- Venlafaxine
- Amitriptyline
- Paroxetine
- Fluoxetine
- Mirtazapine

Your answer is incorrect.

Explanation:

There is limited evidence for efficacy for any pharmacological treatment for akathisia. Options include switching to an antipsychotic with a lower liability for the condition, or adding a beta-adrenergic blocker, a 5-HT2A antagonist (such as mirtazapine or mianserin), or an anticholinergic agent.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 111.

The correct answer is: Mirtazapine

Question 61

Not answered

Marked out of 1.00

Buprenorphine acts on which receptor?

Select one:

- Sigma
- Mu
- Kappa
- Delta
- Alpha

Your answer is incorrect.

Explanation:

Buprenorphine is a partial agonist at mu-opioid receptors.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 478.

The correct answer is: Mu

Question 62

Not answered

Marked out of 1.00

A 40-year-old woman with a past history of psychotic depression is having a relapse but is managed by the Crisis and Home Treatment team in the community. She is currently on lofepramine 180 mg and risperidone 2 mg at bedtime. After a week, she reports to the community worker who visited her at home that her nipples are leaking. She is very concerned and denies she could be pregnant. Which class of medications is most commonly known to cause this problem?

Select one:

- Antidepressants of SSRI class
- Tricyclic antidepressants
- Mood stabilisers
- Antidepressants of MAO inhibitors class
- Neuroleptics

Your answer is incorrect.

Explanation:

When D2 antagonists (neuroleptics) are administered, D2 receptors in the tuberoinfundibular dopamine pathway are blocked. This causes plasma prolactin concentrations to rise, a condition called hyperprolactinaemia. This can be associated with galactorrhea (i.e. breast secretions), sexual dysfunction, gynaecomastia (enlargement of the breasts) in men, and amenorrhea (i.e. irregular or lack of menstrual periods) in women.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 164-165.

The correct answer is: Neuroleptics

Question 63

Not answered

Marked out of 1.00

A 50 year old man is treated for erectile dysfunction using sildenafil as and when required. Availability of which of the following vasodilatory substances is increased by sildenafil?

Select one:

- Nitric oxide
- Endothelin
- Phosphodiesterase
- Angiotensin
- Prostacyclin

Your answer is incorrect.

Explanation:

The discovery in the late 1980s of nitric oxide (NO) as the endothelium-derived relaxing factor was key to understanding the NO-soluble GC (sGC)-cGMP pathway. NO mediates its biological effects by activating sGC and increasing cGMP synthesis which, in turn, activates certain proteins resulting in different actions. cGMP actions are terminated by phosphodiesterase-5 (PDE-5). The synthesis of sildenafil, a PGE-5 inhibitor, and the (serendipity) discovery of its effects on penile erection provided a breakthrough in the treatment of erectile dysfunction and opened new fields of clinical application for this class of drug.

Ref: Andersson KE. PDE5 inhibitors–pharmacology and clinical applications 20 years after sildenafil discovery. British journal of pharmacology. 2018 Jul;175(13):2554-65.

The correct answer is: Nitric oxide

Question 64

Not answered

Marked out of 1.00

Riluzole prevents the release of which neurotransmitter?

Select one:

- Acetylcholine
- Serotonin
- Glutamate
- Norepinephrine
- Dopamine

Your answer is incorrect.

Explanation:

Riluzole was developed to slow the progression of amyotrophic lateral sclerosis (ALS, or Lou Gehrig disease). Theoretically, riluzole binds to voltage-sensitive sodium channels and prevents glutamate release, in an action similar to that proposed for lamotrigine. The idea is that diminishing glutamate release would prevent the postulated excitotoxicity that may be causing the death of motor neurons in ALS.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 352-353.

The correct answer is: Glutamate

Question 65

Not answered

Marked out of 1.00

The mechanism of action of Memantine is

Select one:

- Non competitive NMDA receptor agonist
- Non competitive NMDA receptor antagonist
- Competitive NMDA receptor antagonist
- Competitive NMDA receptor agonist
- Acetylcholinesterase inhibitor

Your answer is incorrect.

Explanation:

The mechanism of memantine involves blockade of N-methyl-d-aspartate (NMDA) glutamate receptors. Unlike ketamine which is a high-affinity noncompetitive blocker, memantine is a non-competitive blocker with low affinity and binds only to actively open NMDA channels. Its receptor dissociation rate is relatively fast, and so it does not accumulate and interfere with normal NMDA activity.

Characteristics of cognitive enhancers:

Characteristic of cognitive enhancer	Donepezil	Rivastigmine	Galantamine	Memantine
Primary mechanism	AChE-I (selective and reversible)	AChE-I (reversible, non-competitive inhibitor)	AChE-I (competitive and reversible)	NMDA receptor antagonist (non-competitive)
Other mechanism	None	BuChE-I	Nicotine receptor agonist	5-HT3 receptor antagonist
Starting dose	5 mg daily	1.5 mg bd (oral) (or 4.6 mg/24 hours patch)	4 mg bd solution or 8 mg XL daily	5 mg daily
Usual treatment dose (and max dose)	10 mg daily	3-6 mg bd (oral) or 9.5 mg/24 hours patch	8-12 mg bd solution or 16-24 mg XL daily	20 mg daily or (10 mg bd)
Adverse effects	Diarrhoea, nausea, and headache most common	Anorexia, dizziness, nausea, vomiting, and diarrhoea most common	Nausea and vomiting most common	Dizziness, headache, fatigue, diarrhoea, and gastric pain most common
Half-life (hours)	~70 (permitting once daily dosing)	~1 (oral) – active for 10 3.4 (patch)	7-8 (oral solution) 8-10 (XL capsules)	60-100
Metabolism	CYP3A4 CYP2D6 (minor)	Minimal involvement of CYP isoenzymes	CYP3A4 CYP2D6	Primarily non-hepatic
Drug-drug interactions	Yes	Interactions unlikely	Yes	Yes
Effect of food on absorption	None	Delays rate and extent of absorption	Delays rate but not extent of absorption	None
Other	Absorbed completely from the GI tract Fairly well-tolerated and widely used	More likely to cause GI adverse effects than donepezil due to more peripheral activity	More likely to cause GI and neuropsychiatric adverse effects than donepezil	Generally better tolerated than cholinesterase inhibitors

AChE-I = acetylcholinesterase inhibitor
BuChE-I = butyrylcholinesterase inhibitor

Adapted from Maudsley Prescribing Guidelines, 2021. pp. 605-6.

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Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 605-606.

The correct answer is: Non competitive NMDA receptor antagonist

Question 66

Not answered

Marked out of 1.00

What is the mechanism of action of atomoxetine when used in the treatment of ADHD?

Select one:

- Dopaminergic agonism
- Noradrenaline reuptake inhibition
- Glutamatergic antagonism
- Serotonin reuptake inhibition
- Dopamine reuptake inhibition

Your answer is incorrect.

Explanation:

Atomoxetine is a selective norepinephrine reuptake inhibitor (NRI). Blocking norepinephrine transporters (NETs) in the prefrontal cortex increases both dopamine and norepinephrine (NE) in the prefrontal cortex and is why NET inhibitors are thought to work in ADHD. However, since there are few NE neurons and NETs in nucleus accumbens, inhibiting NET does not lead to an increase in either NE or DA there and is why NET inhibitors are thought not to have reinforcing, abuse, or addiction potential.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 480.

The correct answer is: Noradrenaline reuptake inhibition

Question 67

Not answered

Marked out of 1.00

Which one among the following properties is not a significant effect of Trazodone?

Select one:

- Serotonin reuptake blockade
- Antihistamine properties
- Strong alpha-1 antagonism
- Norepinephrine reuptake blockade
- Serotonin 2 antagonism

Your answer is incorrect.

Explanation:

Trazodone blocks the serotonin transporter at antidepressant doses. It also has serotonin 2A and 2C, H1 histamine, and alpha-1-adrenergic antagonist properties, which can make it very sedating. At low doses, trazodone does not adequately block serotonin reuptake but retains its other properties.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 535.

The correct answer is: Norepinephrine reuptake blockade

Question 68

Not answered

Marked out of 1.00

A patient who needs a prescription for a new antidepressant is almost certain that he will miss one or two tablets during his treatment due to the nature of his job. What is the most suitable choice of medications?

Select one:

- Escitalopram
- Paroxetine
- Citalopram
- Venlafaxine
- Fluoxetine

Your answer is incorrect.

Explanation:

Due to its longer half-life, missing fluoxetine for a day or two will not usually cause acute withdrawal symptoms. Fluoxetine has a long half-life (2-3 days) and its active metabolite (norfluoxetine) an even longer half-life (2 weeks). Fluoxetine has a long half-life when compared with other SSRIs.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 293.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 562, 742.

The correct answer is: Fluoxetine

Question 69

Not answered

Marked out of 1.00

The effect of which of the following groups of drugs is least influenced by their pharmacokinetics?

Select one:

- SSRIs
- MAO Inhibitors
- Benzodiazepines
- Neuroleptics
- Tricyclics

Your answer is incorrect.

Explanation:

MAOIs such as phenelzine, tranylcypromine, and isocarboxazid are readily absorbed after oral administration and reach peak plasma concentrations with in 2 hours. Whereas their plasma half-lives are in the range of 2-3 hours, their tissue half-lives are considerably longer. They are irreversible enzyme inhibitors, and thus enzyme activity returns only after new enzyme is synthesised about 2-3 weeks later. Thus, the activity of these drugs depends less on pharmacokinetics and more on the synthesis of new MAO to restore normal enzyme activity.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 645.

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 337.

The correct answer is: MAO Inhibitors

Question 70

Not answered

Marked out of 1.00

Which of the following is a cardinal feature of depression?

Select one:

- Anhedonia
- Pessimism
- Suicidality
- Somatisation
- Guilt

Your answer is incorrect.

Explanation:

The two central clinical features of depressive illness are: (1) pervasively depressed and unreactive mood; and 2) anhedonia – the loss of pleasure in previously pleasurable activities.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 63.

The correct answer is: Anhedonia

Question 71

Not answered

Marked out of 1.00

Which of the following clinical features suggests a diagnosis of mania rather than hypomania?

Select one:

- Decreased need for sleep
- Psychomotor agitation
- Inflated self esteem
- Flight of ideas
- Delusions of grandiosity

Your answer is incorrect.

Explanation:

By definition (in both DSM-5 and ICD-11), hypomania is not severe enough to interfere with social or occupational functioning, require admission to hospital, or include psychotic features (delusions or hallucinations).

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 322.

The correct answer is: Delusions of grandiosity

Question 72

Not answered

Marked out of 1.00

Mrs Pinto lost her husband 9 months ago. She had continued attending her bingo sessions and cathedral groups since his death, but recently she has had pervasive guilt and suicidal thoughts. What is the most appropriate diagnosis?

Select one:

- Depression
- Paraphrenia
- Delusional disorder
- Grief
- Dementia

Your answer is incorrect.

Explanation:

Characteristics outside the normal range of grief - that may be indicative of depression – include preoccupation with feelings of worthlessness, thoughts of death, excessive guilt, marked slowing of thoughts and movements, a prolonged period of not being able to function normally, and hallucinatory experiences (other than the image or voice of the deceased).

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 400.

The correct answer is: Depression

Question 73

Not answered

Marked out of 1.00

A 37-year-old woman was admitted to the medical ward with history of sudden onset aphonia. Physical examination and investigations were normal. Her boyfriend has recently invited his mother to stay with them permanently, as his mother is feeling lonely and isolated. What is the most likely diagnosis according to ICD-11 criteria?

Select one:

- Adjustment disorder
- Munchausen syndrome
- Hypochondriasis
- Bodily distress disorder
- Dissociative neurological symptom disorder

Your answer is incorrect.

Dissociative neurological symptom disorder, with speech disturbance is characterised by symptoms such as difficulty with speaking (dysphonia), loss of the ability to speak (aphonia), or difficult or unclear articulation of speech (dysarthria) that are not consistent with a recognised disease of the nervous system, a neurodevelopmental or neurocognitive disorder, other mental, behavioural, or neurodevelopmental disorder, or other medical condition and do not occur exclusively during another dissociative disorder.

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed; ICD-11; World Health Organization, 2020)

The correct answer is: Dissociative neurological symptom disorder

Question 74

Not answered

Marked out of 1.00

A soldier is referred with symptoms of PTSD following return from Iraq. Which of the following feature is not consistent with the diagnosis of PTSD?

Select one:

- He is not able to enjoy watching television
- He is not able to share his war experience with his close friends
- He sleeps in his day time clothes
- He is keen to go to Iraq again
- He has sustained only mild injuries physically

Your answer is incorrect.

Explanation:

Post-traumatic stress disorder is characterised by: **(1) Re-experiencing** the traumatic event(s) in the present in the form of vivid intrusive memories, flashbacks, or nightmares**; (2) Avoidance** of thoughts and memories of the event(s), or avoidance of activities, situations, or people reminiscent of the event(s); **(3) Persistent perceptions of heightened current threat**, for example as indicated by hypervigilance or an enhanced startle reaction to stimuli such as unexpected noises.

Avoidance is a part of PTSD - wanting to go back to Iraq is not consistent with this.

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed; ICD-11; World Health Organization, 2020)

The correct answer is: He is keen to go to Iraq again

Question 75

Not answered

Marked out of 1.00

Mrs Broadbent was diagnosed with compensation neurosis a few decades ago. Which of the following is true regarding this?

Select one:

- Severe difficulties with sleep are noted
- Most patients do not return to work even after settlement
- Higher the severity of injury, greater the symptoms of compensation neurosis
- Absence of frontal headaches
- Improvements tend to occur after financial settlement

Your answer is incorrect.

Compensation neurosis is not an ICD diagnosis; its validity is questionable. Patients show scant improvement even after financial settlement. A follow-up study of 760 litigants found that, of the 264 subjects who were not working at the time of conclusion of litigation and who could be traced, 198 (75%) were not working after an average of 23 months following the finalization of their cases. (Mendelson, 1995. Compensation neurosis revisited. Journal of Psychosomatic Research, 39, pp. 695-706)

The correct answer is: Most patients do not return to work even after settlement

Question 76

Not answered

Marked out of 1.00

An elderly man presents with fluctuation in consciousness and visual hallucinations. His wife reports that he is not able to turn in his bed properly. You suspect

Select one:

- Vascular dementia
- Huntington's disease
- Lewy body dementia
- Depression
- Alzheimer's dementia

Your answer is incorrect.

Explanation:

Not able to turn in bed indicates axial rigidity that may be related to Parkinsonism. His presentation suggests Lewy body dementia.

	Delirium	Alzheimer's Disease (AD)	Lewy Body Dementia (LBD)	Depression
Onset	Sudden	Insidious	Insidious	Recent
Alertness	Fluctuating	Normal except in late phases	Fluctuating	Preserved
Duration	Hours to weeks	Months to years	Months to years	Variable
Orientation	Disorientation with onset	Disorientation occurs late in course	Fluctuating	Intact
Hallucinations	From onset	May occur late in course	From onset; well- formed visual hallucinations	Can occur in depression with psychotic features
Cognitive functioning	Fluctuating with awareness	Progressive deterioration	Progressive deterioration	Initially intact; may deteriorate
Mood	Fluctuating	Labile	Labile	Usually sad
Sundowning	Present	Present	Present	Absent; mood improves as day progresses
Course	Usually reversible with treatment	Irreversible with progressive deterioration	Irreversible with progressive deterioration	Reversible
Adapted from Tampi et al 2019				

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Ref: Tampi RR et al. Psychotic disorders in late life: a narrative review. Therapeutic advances in psychopharmacology. 2019 Oct;9:1-13.

The correct answer is: Lewy body dementia

Question 77

Not answered

Marked out of 1.00

Which of the following is consistent with a diagnosis of bipolar affective disorder, currently hypomanic episode?

Select one:

- Mood incongruent delusions
- 2 weeks duration of symptoms
- Fleeting hallucinations
- Minimal disturbance of psychosocial function
- Mood congruent delusions

Your answer is incorrect.

Explanation:

As defined in ICD-11, a hypomanic episode is a persistent mood state lasting at least several days characterised by mild elevation of mood or increased irritability and increased activity of a subjective experience of increased energy, accompanied by other characteristic symptoms such as rapid speech, rapid or racing thoughts, increased self-esteem, an increase in sexual drive or sociability, decreased need for sleep, distractibility, or impulsive or reckless behaviour. The symptoms are not severe enough to cause marked impairment in occupational functioning or in usual activities or relationships with other, does not necessitate hospitalisation, and there are no accompanying delusions or hallucinations.

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed; ICD-11; World Health Organization, 2020)

The correct answer is: Minimal disturbance of psychosocial function

Question 78

Not answered

Marked out of 1.00

Which of the following features is included in ICD-11 diagnosis for generalised anxiety disorder?

Select one:

- Nightmares
- Panic attacks
- Depression
- Low self confidence
- Muscular tension

Your answer is incorrect.

Explanation:

As defined in ICD-11, generalised anxiety disorder is characterised by marked symptoms of anxiety that persist for at least several months, for more days than not, manifested by either general apprehension (i.e. 'free-floating anxiety') or excessive worry focused on multiple everyday events, most often concerning family, health, finances, and school or work, together with additional symptoms of muscular tension or motor restlessness, sympathetic autonomic over-activity, subjective experience of nervousness, difficulty maintaining concentration, irritability, or sleep disturbance.

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed; ICD-11; World Health Organization, 2020)

The correct answer is: Muscular tension

Question 79

Not answered

Marked out of 1.00

A 33-year-old lady presents with excessive day time sleepiness and auditory hallucinations when waking from sleep. She also recalls episodes of feeling 'pinned down' to her bed even after waking up. The likely diagnosis is

Select one:

- Schizophrenia
- REM behavioural disorder
- Narcolepsy
- Sleep terror
- Primary hypersomnia

Your answer is incorrect.

Explanation:

Classic symptoms of narcolepsy include: (1) Excessive daytime sleepiness; (2) Cataplexy (sudden loss of muscle tone during wakefulness that is evoked by strong, mainly positive, emotions); (3) Sleep paralysis (sometimes up to 10 minutes long); (4) Hypnagogic hallucinations (vivid hallucinations on falling asleep) or hypnopompic hallucinations (hallucinations when the individual is waking up).

The symptoms described here include excessive daytime sleepiness, hypnopompic hallucinations, and sleep paralysis.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 450.

Kornum BR et al. Narcolepsy. Nature reviews Disease primers. 2017 Feb 9;3(1):1-9.

The correct answer is: Narcolepsy

Question 80

Not answered

Marked out of 1.00

Which one of the following is not a negative symptom of schizophrenia?

Select one:

- Alexithymia
- Attentional impairment
- Affective flattening or blunting
- Anhedonia
- Avolition

Your answer is incorrect.

Explanation:

Negative symptoms of schizophrenia include: (1) Flattened affect (limited affective responsiveness); (2) Alogia (poverty of speech); (3) Anhedonia (inability to experience pleasure); (4) Social withdrawal (lack of desire to form relationships); (5) Avolition (lack of motivation); (6) Lethargy; (7) Attentional impairment.

Ref: Barnes TR et al. Evidence-based guidelines for the pharmacological treatment of schizophrenia: Updated recommendations from the British Association for Psychopharmacology. Journal of Psychopharmacology. 2020 Jan;34(1):3-78.

The correct answer is: Alexithymia

Question 81

Not answered

Marked out of 1.00

Simultaneous presence of contrasting attitudes towards a person, an action or an idea is called

Select one:

- Automatic obedience
- Avolition
- Ambivalence
- Anhedonia
- Avoidance

Your answer is incorrect.

Explanation:

Ambivalence is the coexistence of two opposing impulses toward the same thing in the same person at the same time.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 1135.

The correct answer is: Ambivalence

Question 82

Not answered

Marked out of 1.00

Which one of the following terms describes fantasy thinking?

Select one:

- Autistic thinking
- Conceptualism
- Rationalism
- Concrete thinking
- Magical thinking

Your answer is incorrect.

Explanation:

There are three types of thinking: (1) Undirected fantasy thinking (also termed autistic or dereistic thinking) which is quite common, but certain individuals when faced with repeated disappointments or adverse life circumstances may engage in excessive undirected fantasy thinking; (2) Imaginative thinking, which does not go beyond the rational and the possible; (3) Rational thinking or conceptual thinking, which attempts to solve a problem.

Ref: Casey P, Kelly B. Fish's clinical pathology: Signs and symptoms in psychiatry, 4th Edition. 2019. p. 43.

The correct answer is: Autistic thinking

Question 83

Not answered

Marked out of 1.00

A man with epilepsy answers a question correctly, but only after including multiple unnecessary and trivial details. This form of speech is called

Select one:

- Circumstantiality
- Clang association
- Verbigeration
- Tangentiality
- Vorbereiden

Your answer is incorrect.

Explanation:

Circumstantiality occurs when thinking proceeds slowly with many unnecessary and trivial details, but finally the point is reached. The goal of thinking is never completely lost and thinking proceeds towards it by an intricate and convoluted path. Historically, this has been regarded as a feature of the constellation of personality traits occasionally associated with epilepsy. Circumstantiality, however, can also occur in the context of intellectual disability and in individuals with obsessional personality traits, as well as schizophrenia and schizoaffective disorder.

Ref: Casey P, Kelly B. Fish's clinical pathology: Signs and symptoms in psychiatry, 4th Edition. 2019. p. 45.

The correct answer is: Circumstantiality

Question 84

Not answered

Marked out of 1.00

The most common form of synesthesia is

Select one:

- Word taste type
- Colour sound type
- Colour number type
- Touch colour type
- Flavour colour type

Your answer is incorrect.

Explanation:

Synaesthesia is the experience of a stimulus in one sense modality producing a sensory experience in another (e.g. tasting sounds or hearing colours). Over 60 types of synaesthetic phenomena have been described. The apparently most common form (with a 64.4% prevalence among synaesthetes) is grapheme-colour synaesthesia, in which achromatic letters or digits automatically trigger an idiosyncratic colour perceptual experience (e.g., the number '4' induces blue colour percepts).

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 118.

Casey P, Kelly B. Fish's clinical pathology: Signs and symptoms in psychiatry, 4th Edition. 2019. p. 36.

Safran AB, Sanda N. Color synesthesia. Insight into perception, emotion, and consciousness. Current opinion in neurology. 2015 Feb;28(1):36.

The correct answer is: Colour number type

Question 85

Not answered

Marked out of 1.00

Mr. Smith experiences repetitive intrusive images of corpses of his family members in coffins. He had to imagine the same people alive to reduce the anxiety associated with the experience. This compensatory attempt is best described as

Select one:

- Ruminations
- Disaster image
- Obsessional image
- Disruptive image
- Compulsive image

Your answer is incorrect.

Explanation:

In OCD, obsessions are repetitive unwanted thoughts, images, or urges. Compulsions are repetitive behaviours or thoughts that are often, but not always, performed to address anxiety associated with obsessions. The drive to perform a compulsion is recognised by the patient as their own (i.e. there is not sense of 'possession' or passivity), but it is associated with a subjective sense of need to perform it, often in order to avoid the occurrence of an adverse event. The patient may resist carrying out the compulsion for a time, at the expense of mounting anxiety.

Two types of recurrent images may be described in OCD. Obsessive image refers to recurrent images that are recognised as irrelevant, senseless appearing in one's own mind but with no willful control. Sometimes patients may voluntarily produce images that are counter-intuitive in order to negate the feared effects of such obsessive images - these are called as compulsive images.

Ref: Hirschtritt ME, Bloch MH, Mathews CA. Obsessive-compulsive disorder: advances in diagnosis and treatment. Jama. 2017 Apr 4;317(13):1358-67.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 104.

The correct answer is: Compulsive image

Question 86

Not answered

Marked out of 1.00

On a nice sunny day you are driving along a country road where you have never been before. You suddenly have a sense of familiarity for the place. This is known as:

Select one:

- Transient amnesia
- Derealisation
- Reduplicative paramnesia
- Jamais vu
- Deja vu

Your answer is incorrect.

Explanation:

Déjà vu is a sense that events being experienced for the first time have been experienced before. Jamais vu is the sensation that events or situations are unfamiliar, although they have been experienced before. Both déjà vu and jamais vu can be everyday experiences, but also non-specific symptoms of a number of disorders, including temporal lobe epilepsy, schizophrenia, and anxiety disorders.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 105, 113.

The correct answer is: Deja vu

Question 87

Not answered

Marked out of 1.00

A 49-year-old lady repeatedly presents to police asking them to arrest her for a murder despite never having committed this crime. This condition is known as:

Select one:

- Anhedonia
- Delusion of guilt
- Delusion of reference
- Depersonalisation
- Nihilistic delusion

Your answer is incorrect.

Explanation:

A delusion of guilt is a delusional belief that one has committed a crime or other reprehensible act. It can be a feature of psychotic depressive illness.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 107.

The correct answer is: Delusion of guilt

Question 88

Not answered

Marked out of 1.00

"Today I took a train from Derby to Matlock with Jeff, my dog. I carried my knife and pepper spray as we walk through the town, as I am convinced someone will abduct and mutilate me and kill Jeff". This can be described as a

Select one:

- Nihilistic delusion
- Delusion of misidentification
- Delusion of persecution
- Delusion of reference
- Delusion of control

Your answer is incorrect.

Explanation:

A persecutory delusion is a delusional belief that one's life is being interfered with in a harmful way. Some patients believe that they or their loved one are about to be killed, or are being tortured.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 115.

Casey P, Kelly B. Fish's clinical pathology: Signs and symptoms in psychiatry, 4th Edition. 2019. p. 50.

The correct answer is: Delusion of persecution

Question 89

Not answered

Marked out of 1.00

A 45-year-old woman in a psychiatric inpatient unit complains that someone is injecting blood into her body on a regular basis. On further questioning she points to a black mole in her hand as an explanation for her belief. This is best termed as

Select one:

- Delusional perception
- Delusional elaboration
- Persecutory delusion
- Somatic hallucination
- Delusional infestation

Your answer is incorrect.

Explanation:

A delusional elaboration is a secondary delusion which arises in a manner which is understandable as the patient attempting to find explanations for primary psychopathological processes (e.g. a patient with persistent auditory hallucinations developing a belief that a transmitter has been placed in their ear).

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 106.

The correct answer is: Delusional elaboration

Question 90

Not answered

Marked out of 1.00

Which of the following attachment behaviour pattern has been commonly reported in children subjected to abuse?

Select one:

- Anxious/avoidant attachment
- Secure attachment
- Multiple attachment
- Disorganised attachment
- Ambivalent/resistant attachment

Your answer is incorrect.

Explanation:

Mary Ainsworth devised the Strange Situation procedure with 12- to 18-month infants, which categorised infant-parent relationships into three distinct groups: secure, insecure avoidant, and insecure ambivalent. The procedure involves separations and reunions with caregivers and observes response. A fourth category, disorganised, was added by Mary Main in 1986. Bizarre, confused, or contradictory behaviour is characteristic of disorganised attachment behaviour. It is often associated with maltreatment/abuse and/or parental trauma.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 658-659.

The correct answer is: Disorganised attachment

Question 91

Not answered

Marked out of 1.00

Fears that show no particular age trend include:

Select one:

- Fear of open spaces
- Fear of snakes
- Fear of the dark
- Fear of animals
- Fear of sex

Your answer is incorrect.

Explanation:

Fears that show no particular age trend include fear of snakes or storms. While babies and very young children do not usually fear snakes, they are unusually skilled at detecting them and show a predisposition to learn to fear snakes if they have bad experiences or even if they are exposed to negative portrayals of them in the media.

Souchet J, Aubret F. Revisiting the fear of snakes in children: the role of aposematic signalling. Scientific reports. 2016 Nov 25;6(1):1-7.

Age	Developmental fear (onset age of pathological *phobias* in italics)
3 years	Fear of animals
4 or 5 years	Fear of the dark
5+ years	Fear of imaginary creatures
Around 7 years	Onset for animal *phobias*
Around 8 years	Blood/injection/injury *phobias*
Around 5-10 years	Natural environment *phobias*
Late childhood/adult life	Fear of sex; open spaces
Teenage onwards	Failure, illness, and death
Around 20 years	Situational *phobias*
No age trend	Fear of snakes or storms



The correct answer is: Fear of snakes

Question 92

Not answered

Marked out of 1.00

Deductive reasoning is achieved in which stage of Piaget's cognitive development theory?

Select one:

- Operational stage
- Concrete operational stage
- Pre-operational stage
- Sensorimotor stage
- Formal operational stage

Your answer is incorrect.

Explanation:

According to Piaget, the stage of formal operations lasts from about 11 years of age through the end of adolescence. The stage is so named because young persons' thinking operates in a formal, highly logical, systematic, and symbolic manner. It is characterised by the ability to think abstractly, to reason deductively, and to define concepts, and also by the emergence of skills for dealing with permutations and combinations. Adolescents attempt to deal with all possible relations and hypotheses to explain data and events during this stage. Hypotheticodeductive thinking, the highest organisation of cognition, enables people to make a hypothesis or proposition and to test it against reality. Deductive reasoning moves from the general to the particular and is more complicated process than inductive reasoning, which moves from the particular to the general.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 1009.

Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Formal operational stage

Question 93

Not answered

Marked out of 1.00

An 8-year-old boy is most likely to fall under which stage of Erikson's psychosocial development?

Select one:

- Trust vs mistrust
- Identity vs role confusion
- Initiative vs guilt
- Autonomy vs shame
- Industry vs inferiority

Your answer is incorrect.

Explanation:

Erik Erikson created an original and highly influential theory of psychological development and crisis occurring in periods that extended across the entire life cycle. Stage 4 takes place from about 5 years to about 13 years. Its focus is industry versus inferiority. The child develops industry by learning new skills and takes pride in the things made. A child unprepared for this stage of psychosocial development, either through an insufficient resolution of previous stages or by current interference, may develop a sense of inferiority and inadequacy.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 1071-1074.

The correct answer is: Industry vs inferiority

Question 94

Not answered

Marked out of 1.00

In Ainsworth's strange situation experiment, a child showed no signs of distress at leaving the mother and ignored her on return. Which type of attachment behaviour is this?

Select one:

- Disorganized attachment behaviour
- Insecurely attached anxious ambivalent behaviour
- Insecurely attached anxious resistant behaviour
- Insecurely attached anxious avoidant behaviour
- Securely attachment behaviour

Your answer is incorrect.

Explanation:

Mary Ainsworth devised the Strange Situation procedure with 12- to 18-month infants, which categorised infant-parent relationships. The procedure involves separations and reunions with caregivers and observes response. In insecure-avoidant attachment, the child seems unconcerned at separation or reunion. It is associated with unresponsive parenting.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 658-659.

The correct answer is: Insecurely attached anxious avoidant behaviour

Question 95

Not answered

Marked out of 1.00

Which one of the following is associated with classifying personality according to body build?

Select one:

- Kretschmer
- Rutter
- Kraeplin
- Russell
- Fairburn

Your answer is incorrect.

Explanation:

Ernst Kretschmer theorised that there are four basic body types, each associated with its own distinctive pattern of personality characteristics: (1) Aesthetic: slender, long-limbed; (2) Athletic: muscular, well-built; (3) Dysplastic: misshapen, ugly; (4) Pyknic: short, squat, rounded.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Kretschmer

Question 96

Not answered

Marked out of 1.00

Which type of learning behaviour occurs without apparent reward or reinforcement?

Select one:

- Modelling
- Observational learning
- Insight learning
- Latent learning
- None of the listed options

Your answer is incorrect.

Explanation:

Latent learning is animal learning that occurs without any apparent reinforcement or reward, as when a rat explores a maze and is later shown to have learned to find its way around it. In human learning, the phenomenon is called incidental learning.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Latent learning

Question 97

Not answered

Marked out of 1.00

Which one of the following statements about memory loss following ECT is incorrect?

Select one:

- Memory impairment is more pronounced with unilateral ECT
- The period closest to receiving ECT is the least well-remembered in patients with ECT-related memory issues
- A third of patients report persistent memory loss in some surveys
- There may be both anterograde and retrograde amnesia
- Most studies did not measure autobiographical memory over the long term

Your answer is incorrect.

Explanation:

Memory loss following ECT is usually temporary, and both anterograde and retrograde amnesia are reported. A third of patients may report persistent memory loss following ECT (Rose et al., 2003). Research has focused on retrograde amnesia because of highly publicised claims that ECT causes more enduring deficits in past memories (especially autobiographical) than new memories. Research shows that the period closest to receiving ECT is least well-remembered and can be permanently lost. Autobiographical loss does occur and is related to how ECT is administered. Recommendations to minimise memory loss include: use of right unilateral ECT; brief pulse, rather than sine wave, ECT; dose titration; and limited number and frequency of ECT sessions.

Ref: Rose D, Wykes T, Leese M, *et al.* Patients' perspectives on electroconvulsive therapy: systematic review. BMJ 2003;326:1363.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 308.

The correct answer is: Memory impairment is more pronounced with unilateral ECT

Question 98

Not answered

Marked out of 1.00

Which of the following instruments assess personality as a construct spread over ten clinical scales?

Select one:

- International personality disorders examination
- Eysenck's personality questionnaire
- NEO personality inventory
- Thematic appreception test
- Minnesota multiphasic personality inventory

Your answer is incorrect.

Explanation:

The Minnesota Multiphasic Personality Inventory (MMPI) is a self-report questionnaire consisting of 567 questions covering eight areas of psychopathology and two areas of personality type. The scales are: Hypochondriasis, Depression, Hysteria, Psychopathic deviate, Masculinity-femininity, Paranoia, Psychasthenia, Schizophrenia, Hypomania, and Social introversion. There are also four additional validity (test-taking attitude) scales.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 96.

Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Minnesota multiphasic personality inventory

Question 99

Not answered

Marked out of 1.00

Which one among the following is an objective personality test?

Select one:

- Rorschach ink blot test
- Sentence completion test
- Draw a person test
- Thematic apperception test
- Minnesota multiphasic personality inventory

Your answer is incorrect.

Explanation:

The Minnesota Multiphasic Personality Inventory (MMPI) is a self-report questionnaire consisting of 567 questions covering eight areas of psychopathology and two areas of personality type. The scales are: Hypochondriasis, Depression, Hysteria, Psychopathic deviate, Masculinity-femininity, Paranoia, Psychasthenia, Schizophrenia, Hypomania, and Social introversion. There are also four additional validity (test-taking attitude) scales.

Projective (non-objective) tests of personality include the Rorschach test (10 cards), Thematic Apperception Test (TAT-with pictures of persons and scenes), Sentence Completion Test, Holtzman Inkblot Technique (TIT), Figure drawing (typically human forms but can involve houses or other forms e.g. Draw-a-Person-Test or DAPT) and Make-a-Picture Story (MAPS).

Cattell 16 PF scale is an objective personality test devised from a nonclinical population. It includes a lie scale, frequency and correction scales.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 96

Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 59.

The correct answer is: Minnesota multiphasic personality inventory

Question 100

Not answered

Marked out of 1.00

Which one among the following is a feature of attentional processing according to Shiffrin and Schneider's divided attention theory?

Select one:

- Automatic through extensive training
- Not limited
- Needs more mental effort.
- Unconscious
- Resistant to change.

Your answer is incorrect.

According to Shiffrin and Schneider's divided attention theory (1977), automatic processing is unconscious, not limited, automatic through extensive training, and once learnt are resistant to change. But Attentional (controlled) processing is consciously directed, needs focussed attention and more mental effort and also has limited capacity.

The correct answer is: Needs more mental effort.

Question 101

Not answered

Marked out of 1.00

Which of the following offers a psychological explanation regarding the aetiology of OCD?

Select one:

- Reciprocal inhibition
- Punishment
- Negative reinforcement
- Positive reinforcement
- Reward orientation

Your answer is incorrect.

Explanation:

Psychologically, obsessions are conditioned (neutral) stimuli, associated with an anxiety-provoking event. Compulsions are learnt and negatively reinforced, as they are a form of anxiety-reducing avoidance.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 385.

The correct answer is: Negative reinforcement

Question 102

Not answered

Marked out of 1.00

Star chart is based on the principles of

Select one:

- Avoidance conditioning
- Classical conditioning
- Operant conditioning
- Higher order conditioning
- Trace conditioning

Your answer is incorrect.

Explanation:

Behaviour modification is based on operant conditioning. Forms of behavioural modification include the use of secondary reinforcement, such as 'token economy', in which socially desirable/acceptable behaviours are rewarded with tokens that can be exchanged for other material items or privileges, or 'star charts' where children's good behaviour is rewarded when a certain level is achieved.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 908.

The correct answer is: Operant conditioning

Question 103

Not answered

Marked out of 1.00

Which part of the brain is involved in mediating performances in short term memory?

Select one:

- Left and right ventricles
- Parietal lobes
- Prefrontal lobes
- Entorhinal cortex of the medial temporal lobe
- Limbic lobe

Your answer is incorrect.

Brain imaging and neuropsychological studies provide strong evidence that a.The brain areas mediating performances in STM are principally the pre-frontal lobes b.The phonological STM system is mediated by the left hemisphere regions of Broca's' area and prefrontal cortex. c.The visuospatial STM system is mediated by the parietal and prefrontal areas of the right hemisphere. d.The brain areas responsible for LTM includes the regions of the limbic system especially the hippocampus and the Entorhinal cortex of the medial temporal lobe.

The correct answer is: Prefrontal lobes

Question 104

Not answered

Marked out of 1.00

A candidate who has appeared for a mock exam before the actual exam does better than a candidate who has not. This is called

Select one:

- Regression of mean
- Hawthorne effect
- Observer effect
- Halo effect
- Practice effect

Your answer is incorrect.

Practice effects occur when a participant in an experiment performs a task after having performed the same task earlier. Subjects can either have a positive (subjects become better at performing the task) or negative (subjects become worse at performing the task) effect. Repeated measures designs are almost always affected by practice effects.

The correct answer is: Practice effect

Question 105

Not answered

Marked out of 1.00

The most common effect of cognitive dissonance is

Select one:

- Psychological distress
- Cognitive distortions
- Memory disturbances
- Formal thought disorder
- Behavioural modification

Your answer is incorrect.

Explanation:

Cognitive dissonance is a motivating state of tension or distress that tends to generate three types of dissonance-reducing behaviour: changing one of the cognitions, decreasing the perceived importance of the dissonant cognitions, and/or adding further (justifying) cognitions (the easiest and commonest method of dissonance reduction).

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Psychological distress

Question 106

Not answered

Marked out of 1.00

A Malaysian man shows sudden excitement with marked violence after a period of withdrawal. He returns to premorbid state after a violent incident. This is called as

Select one:

- Latah
- Cathard
- Windigo
- Amok
- Piblokto

Your answer is incorrect.

Explanation:

Amok is a dissociative episode characterised by a period of depression followed by an outburst of violent, aggressive, or homicidal behaviour. Episodes tend to be caused by a perceived insult and are often accompanied by persecutory ideas, automation, amnesia, and exhaustion. Patients return to premorbid states following the episode. Amok seems to be prevalent only among males. The term originated in Malaysia.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 1053.

The correct answer is: Amok

Question 107

Not answered

Marked out of 1.00

The following is true about the prima facie (main) principles of medical ethics:

Select one:

- Virtue theory: principle based on the characteristics of moral agents
- Autonomy: Respecting patient's views despite disagreement
- Utilitarianism: principle based on the greatest benefit for the greater number of people
- Deontology: principle based on the act as stemming from duties irrespective of the moral agent characteristics
- Teleology: principle based on consequences of the act

Your answer is incorrect.

Explanation:

The four principles of Beauchamp and Childress – autonomy, non-maleficence, beneficence, and justice – have been extremely influential in the field of medical ethics. Autonomy requires that a person act intentionally after being given sufficient information and time to understand the benefits, risks, and costs of all options.

Note that even though the pairs are correct with respect to the explanations attached, except autonomy the other ethical principles stated here belong to 'higher-order' values.

Ref: Page K. The four principles: Can they be measured and do they predict ethical decision making?. BMC medical ethics. 2012 Dec;13(1):1-8.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 829.

The correct answer is: Autonomy: Respecting patient's views despite disagreement

Question 108

Not answered

Marked out of 1.00

Which of the following is most associated with an increased risk of developing schizophrenia?

Select one:

- Being frequently placed in a 'double-bind situation' as a child
- Being born in an urban area
- Maternal cannabis use during pregnancy
- Experiencing high expressed emotion in the family home
- Maternal alcohol use during pregnancy

Your answer is incorrect.

Explanation:

Urbanisation at birth has been associated with an increased risk of schizophrenia.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 191.

The correct answer is: Being born in an urban area

Question 109

Not answered

Marked out of 1.00

When a psychiatric patient is unwell, family members who advocate for admission to a mental hospital are unlikely to provide a realistic picture of what the patient will experience. After hospitalisation, the patient may feel 'conned' by them. What was the term used by Goffman to describe this?

Select one:

- Institutional neurosis
- Role stripping
- Mortification
- Batch living
- Betrayal funnel

Your answer is incorrect.

Explanation:

Erving Goffman was a Canadian-born US sociologist whose writing provides insights into the mental hospital environment, particularly as experienced by patients. He suggested that family who advocate for admission to a mental hospital are unlikely to provide a realistic picture of what the patient will experience. The patient will be told that they will get required medical treatment and a rest, and then be released. In some cases, these advocates of admission to a mental hospital may be concealing what they know, but more likely they will be telling what they believe to be the truth. Very quickly, though, the patient experiences the hospital very differently, and realizes that the information that was given to them about life in the hospital has had the effect of reducing the resistance to entering it. Thus, the patient may feel 'conned', and these significant figures can function as part of a betrayal funnel.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

Capps D. The mortification of the self: Erving Goffman's analysis of the mental hospital. Pastoral psychology. 2016 Feb 1;65(1):103-26.

The correct answer is: Betrayal funnel

Question 110

Not answered

Marked out of 1.00

Of the following, the life event with the highest rating on the Holmes and Rahe scale is

Select one:

- Marriage
- Childbirth
- Divorce
- Job loss
- Jail term

Your answer is incorrect.

Explanation:

In 1967, psychiatrists Thomas Holmes and Richard Rahe examined over 5000 patients' medical records to determine whether stressful events cause illnesses. Patients ranked a list of 43 life events based on a relative score. Each event had a different 'weight' for stress. Their results were published as the Social Readjustment Rating Scale, now commonly known as the Holmes and Rahe Stress Scale. The scale lists common stressful events and arbitrarily assigns a value of 50 Life Change Units (LCUs) to the stress caused by marriage. Death of a spouse has the highest LCUs at 100, followed by divorce (73 LCUs), and marital separation (65 LCUs).

Ref: Noone PA. The Holmes–Rahe Stress Inventory. Occupational Medicine. 2017 Oct 1;67(7):581-2.

Holmes TH, Rahe RH. The social readjustment rating scale. Journal of psychosomatic research. 1967.

The correct answer is: Divorce

Question 111

Not answered

Marked out of 1.00

Which of the following research materials is covered by the Helsinki Declaration?

Select one:

- Identifiable human data
- Anonymised human data
- Animal data
- Environmental data
- Non-clinical digital data

Your answer is incorrect.

Explanation:

The World Medical Association (WMA) has developed the Declaration of Helsinki as a statement of ethical principles for medical research involving human subjects, including research on identifiable human material and data. Adopted by the WMA's General Assembly in 1964, it is the most important set of guidelines about research on human participants. It both reflects and shapes the ethos of international research ethics. It is a living instrument and is reviewed and revised regularly. Its latest revision was in 2013.

Ref: https://www.wma.net/policies-post/wma-declaration-of-helsinki-ethical-principles-for-medical-research-involving-human-subjects/

Malik AY, Foster C. The revised Declaration of Helsinki: cosmetic or real change?. Journal of the Royal Society of Medicine. 2016 May;109(5):184-9.

The correct answer is: Identifiable human data

Question 112

Not answered

Marked out of 1.00

A man who works full time at the local council office has headache, abdominal pain and dizziness. He feels tired and has a fever. Which of the following best describes the above situation?

Select one:

- Health behaviour
- Illness
- Sick role
- Abnormal illness behaviour
- Illness behaviour

Your answer is incorrect.

The description given here fits the illness itself rather than the patient's response or social changes that take place.

The correct answer is: Illness

Question 113

Not answered

Marked out of 1.00

A 45 year old man holds out his arm to allow measurement of blood pressure as soon as the staff nurse walks towards him following the daily routine of medication dispensing. Which of the following is true?

Select one:

- As the patient is competent consent is not needed
- Explicit consent is not required as this is a situation of necessity
- Explicit consent need not be sought for routine procedures
- Blood pressure is measured under principle of best interests
- Implied consent can be assumed

Your answer is incorrect.

Explanation:

Consent may be implied (i.e. the patient does not object to, and cooperates with, the procedure) or may be express (i.e. oral or written permission is explicitly asked for and recorded, often on a detailed consent form). Generally, express (or explicit) consent is obtained for non-trivial or invasive procedures, and for some interventions (e.g. operations), it is mandatory. In contrast, implied consent is assumed here.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 937.

The correct answer is: Implied consent can be assumed

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